

COMP 4423: IMAGE RETRIEVAL

Easy Computer Vision

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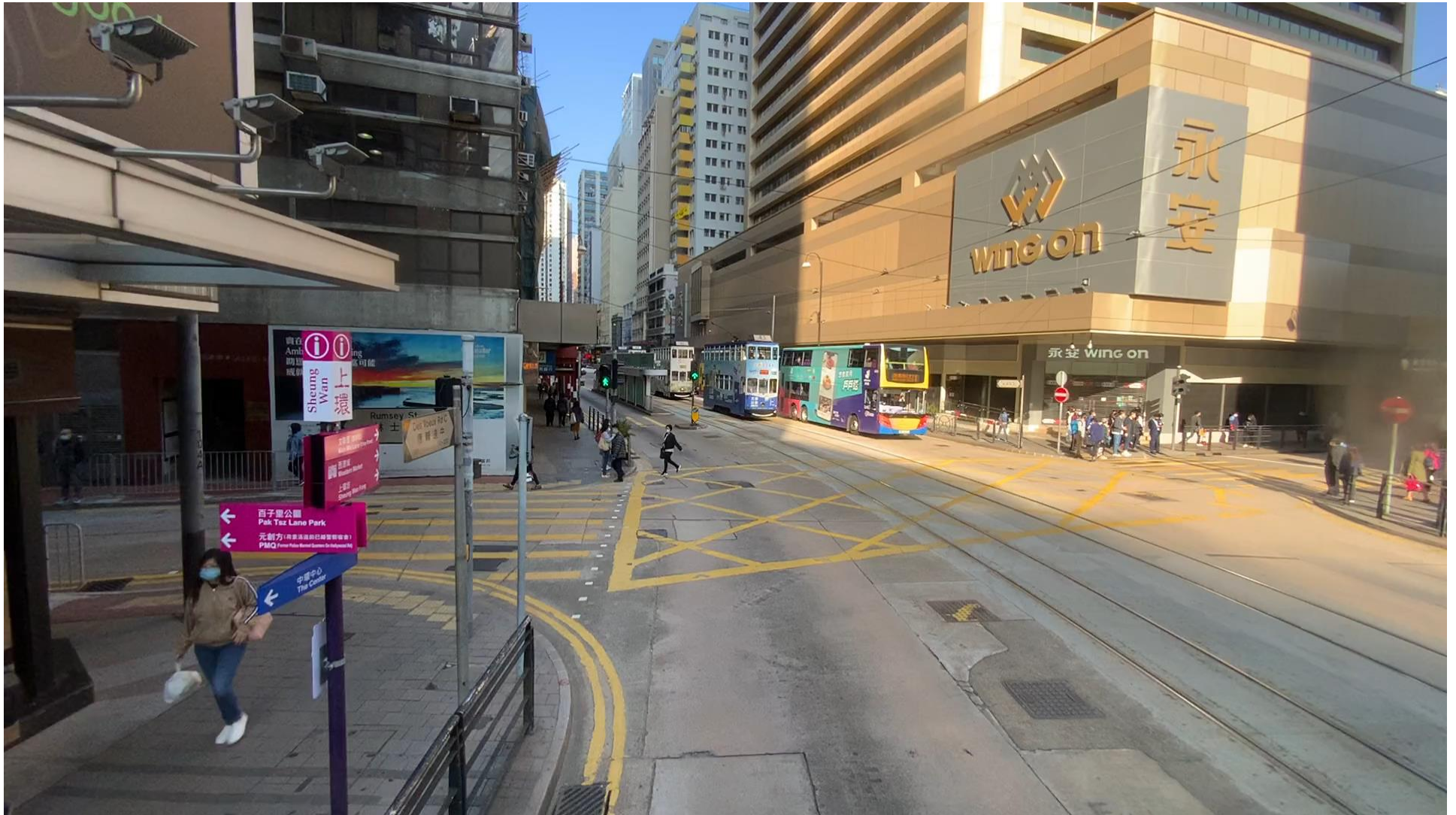


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香港理工大學

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Original Video



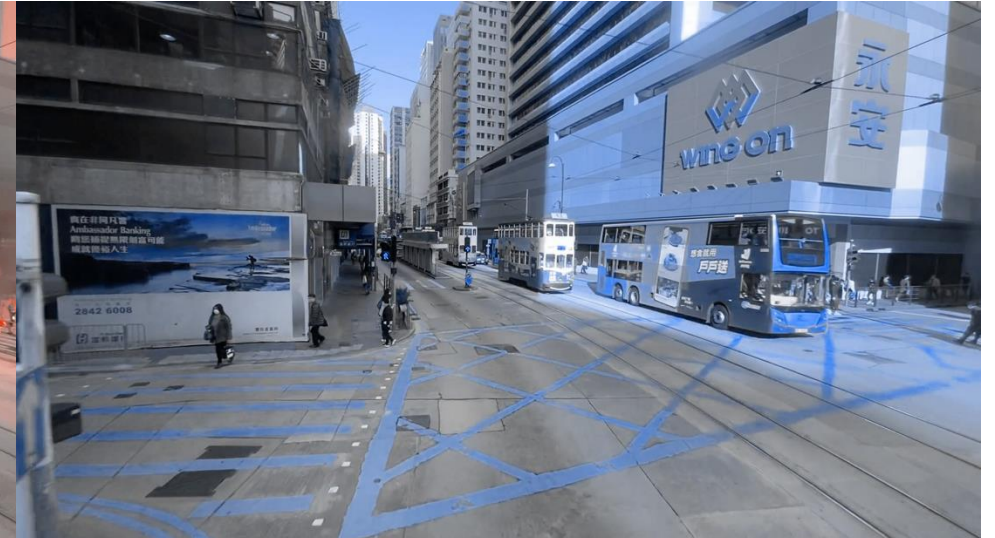


Modified Video





New Toy





Outline

- Clustering
- K-Means
- Content-based image retrieval (CBIR)
- Bag of Visual Words (BoVW)



In Feature Extraction, we teach the computers to represent the “content” of the images.

How can we use these “content”?



Recall where we started

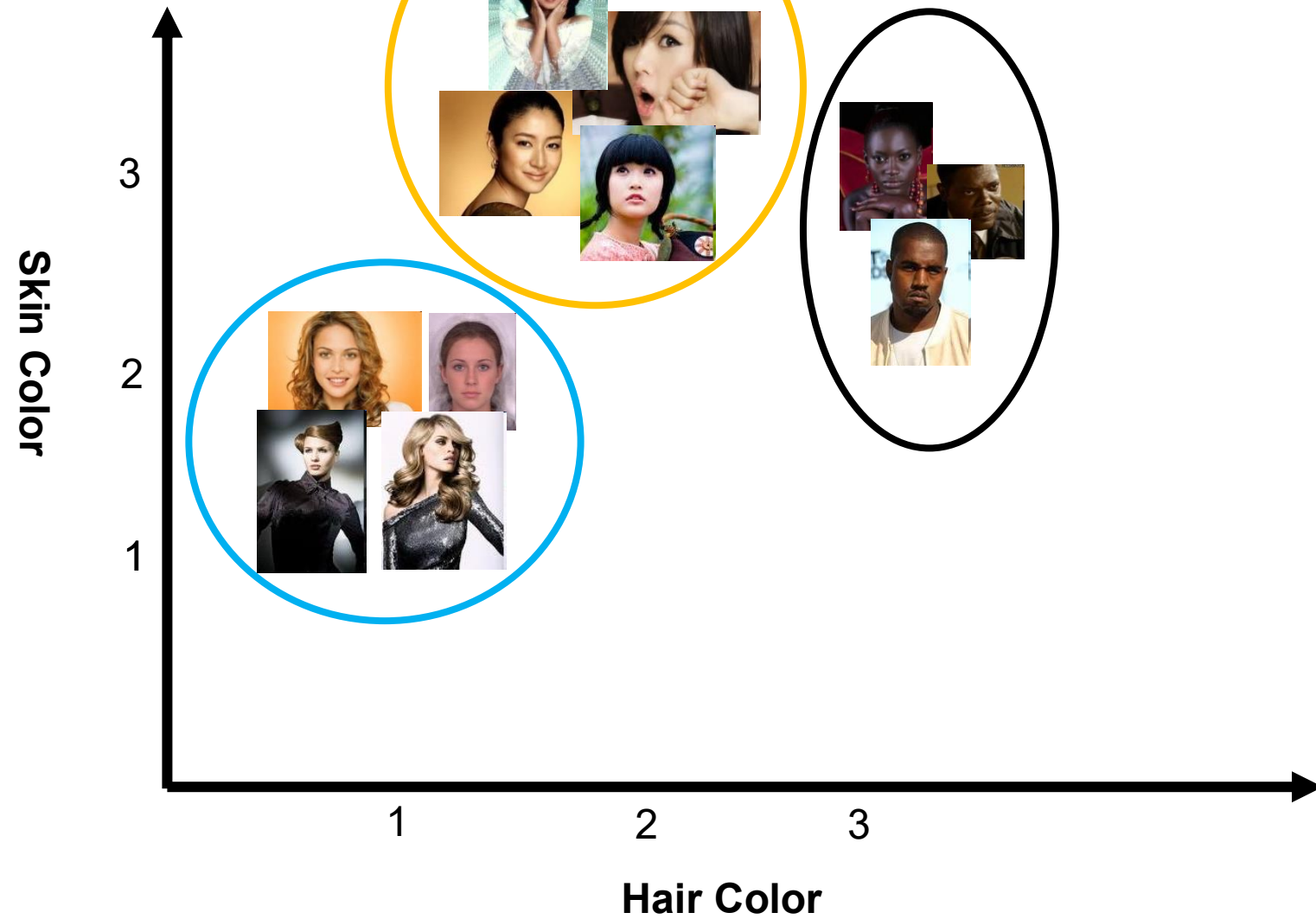


How do you group them?





Feature Space





Images become “numbers” (feature vectors) in the **feature space** after the feature extraction.

To use the “content” is to play those “numbers”. This holds to all we’re going to learn in the rest of this course.

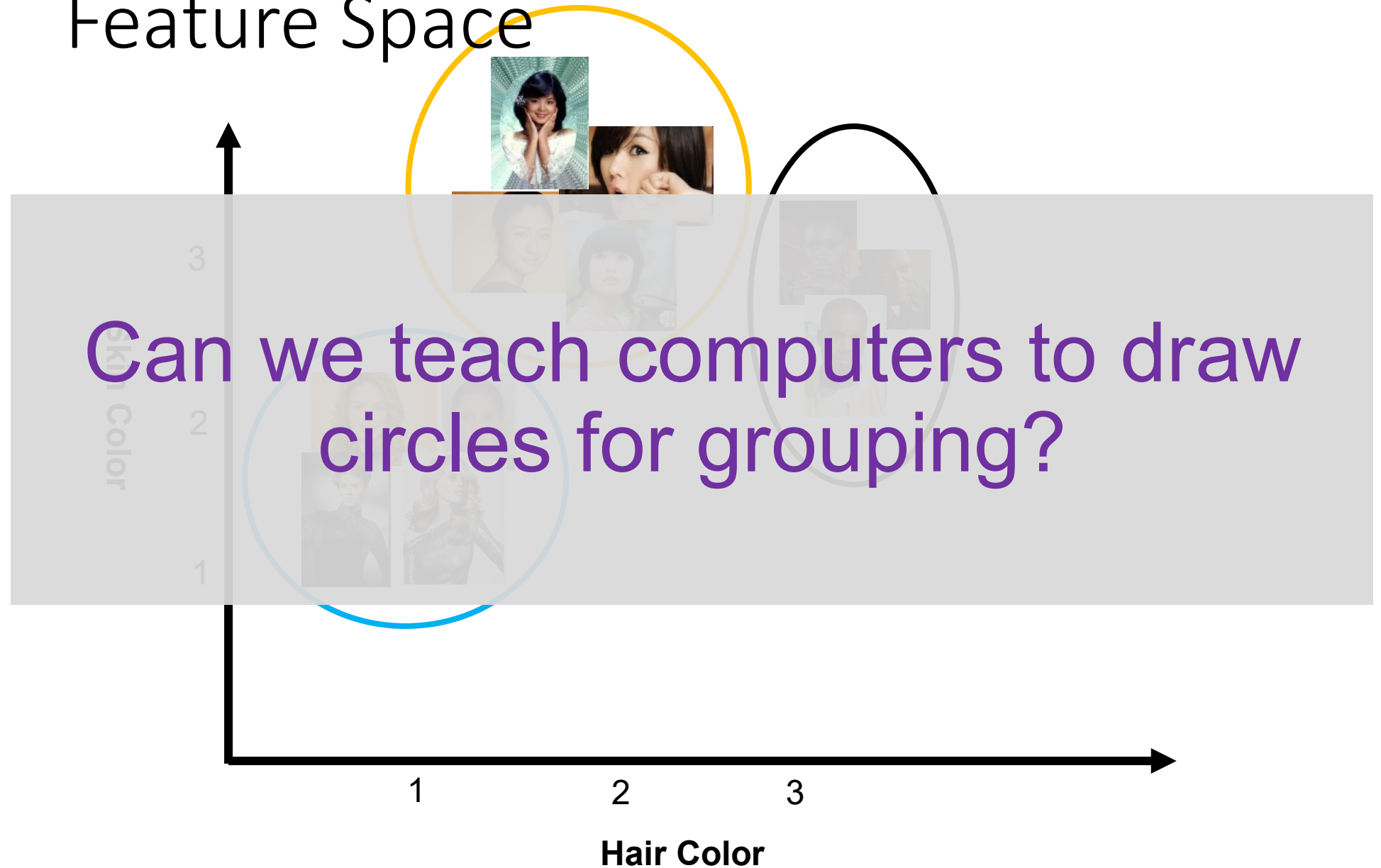


In a more general sense, this applies to texts, audios, videos, and a wide range of other media/information.

We're all trying to represent things as "numbers" in the feature space, making them "readable" for computers.



Feature Space





We can answer the question now.

To group the images is to make the circles as far as possible from each other, while the images inside the same circle as close as possible.

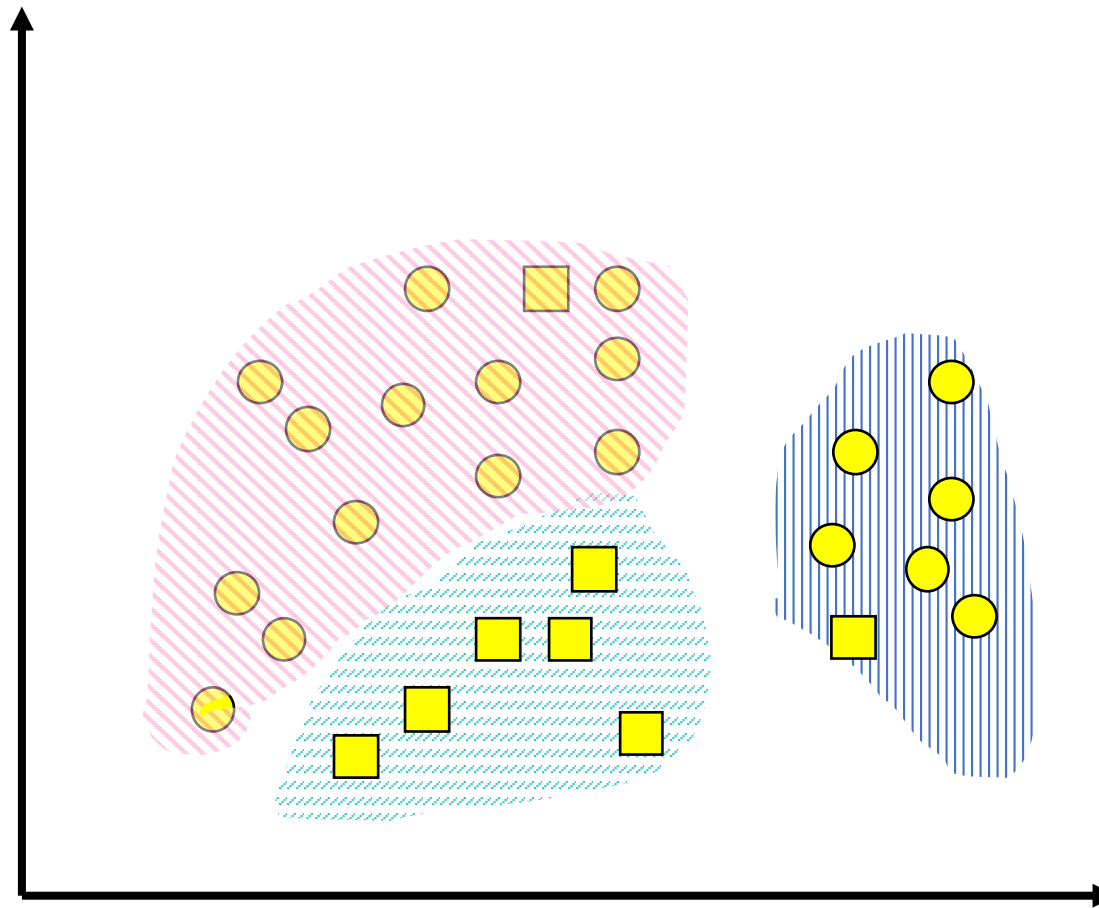


This is the idea of **clustering**: to maximize the **inter-cluster** distance while to minimize the **intra-cluster** distance.





Think in the feature space





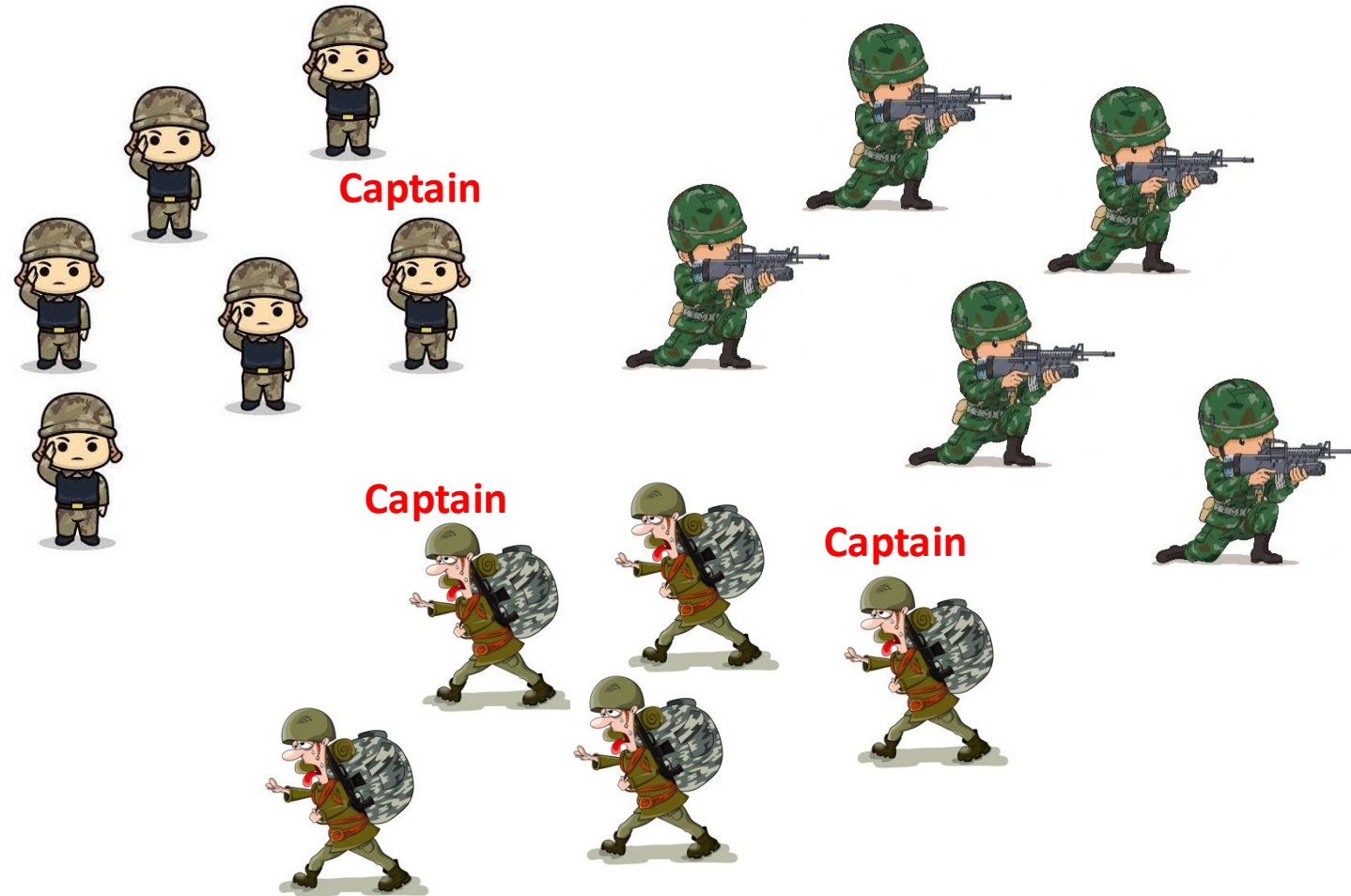
Think in the feature space

> K-Means

- 1: Pick a number (K) of cluster centers (at random)
- 2: Assign every item to its nearest cluster center (e.g. using Euclidean distance)
- 3: Move each cluster center to the **mean** of its assigned items
- 4: Repeat steps 2,3 until convergence (change in cluster assignments less than a threshold)

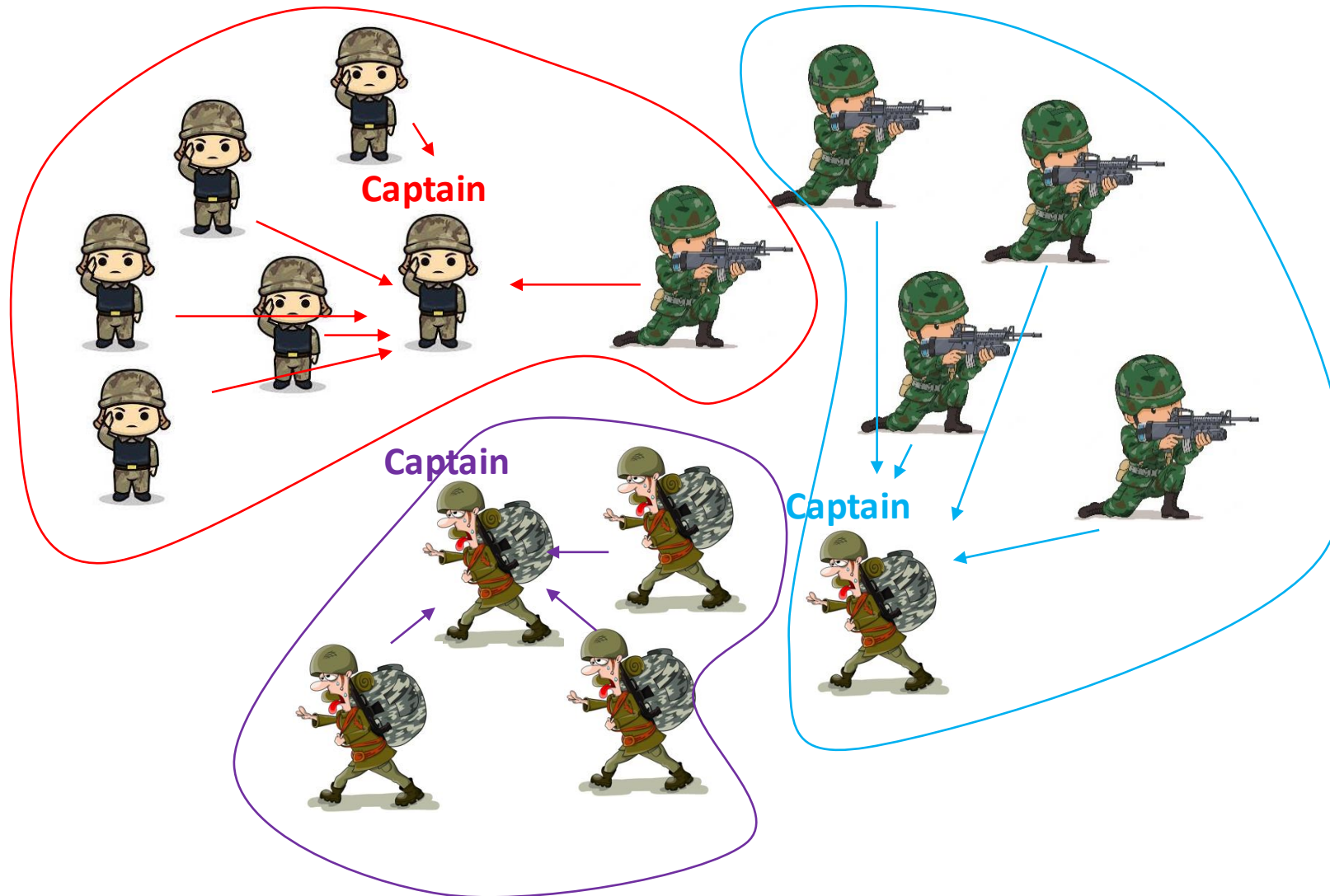


1. Captains assigned randomly



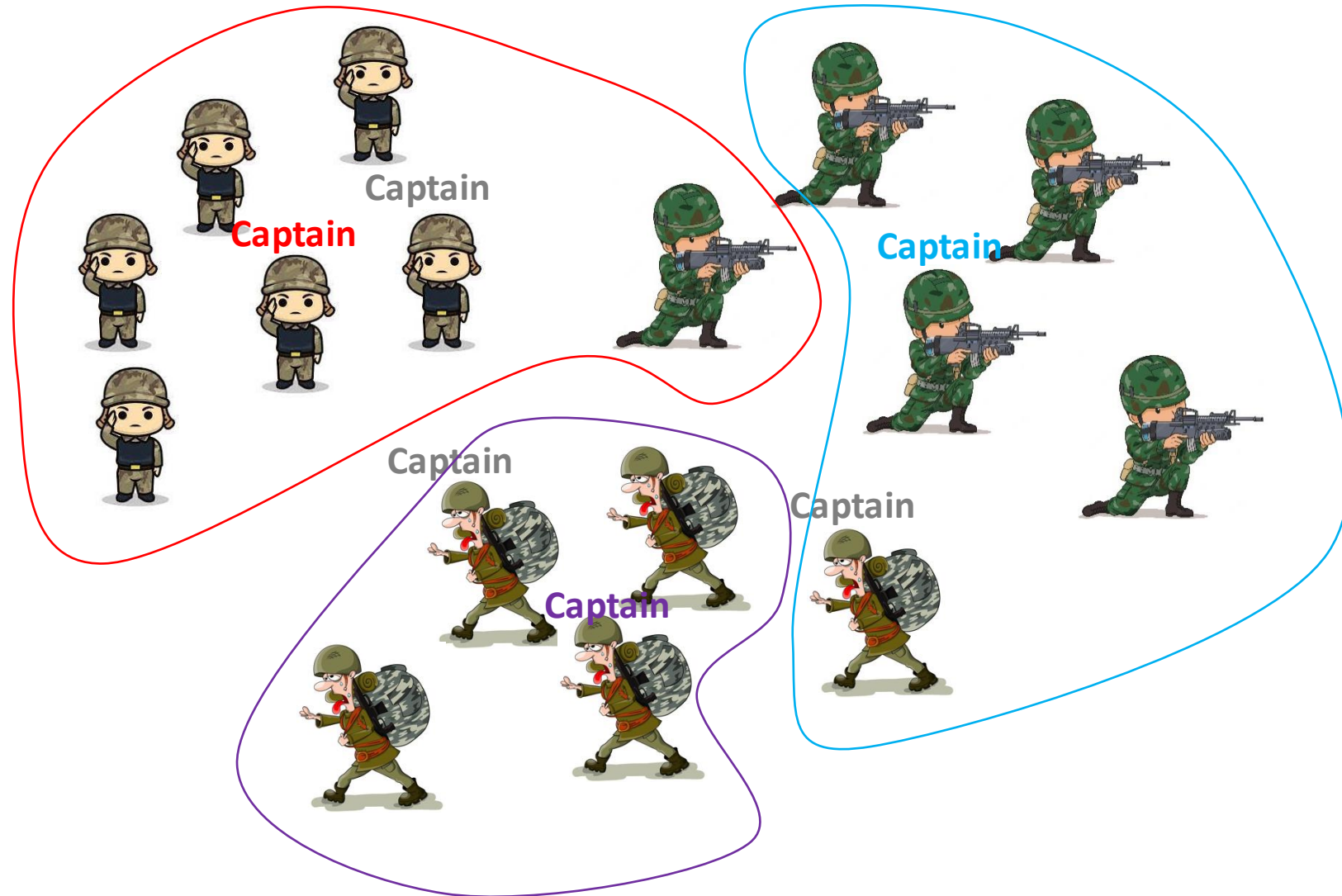


2. Soldiers report to the nearest captains



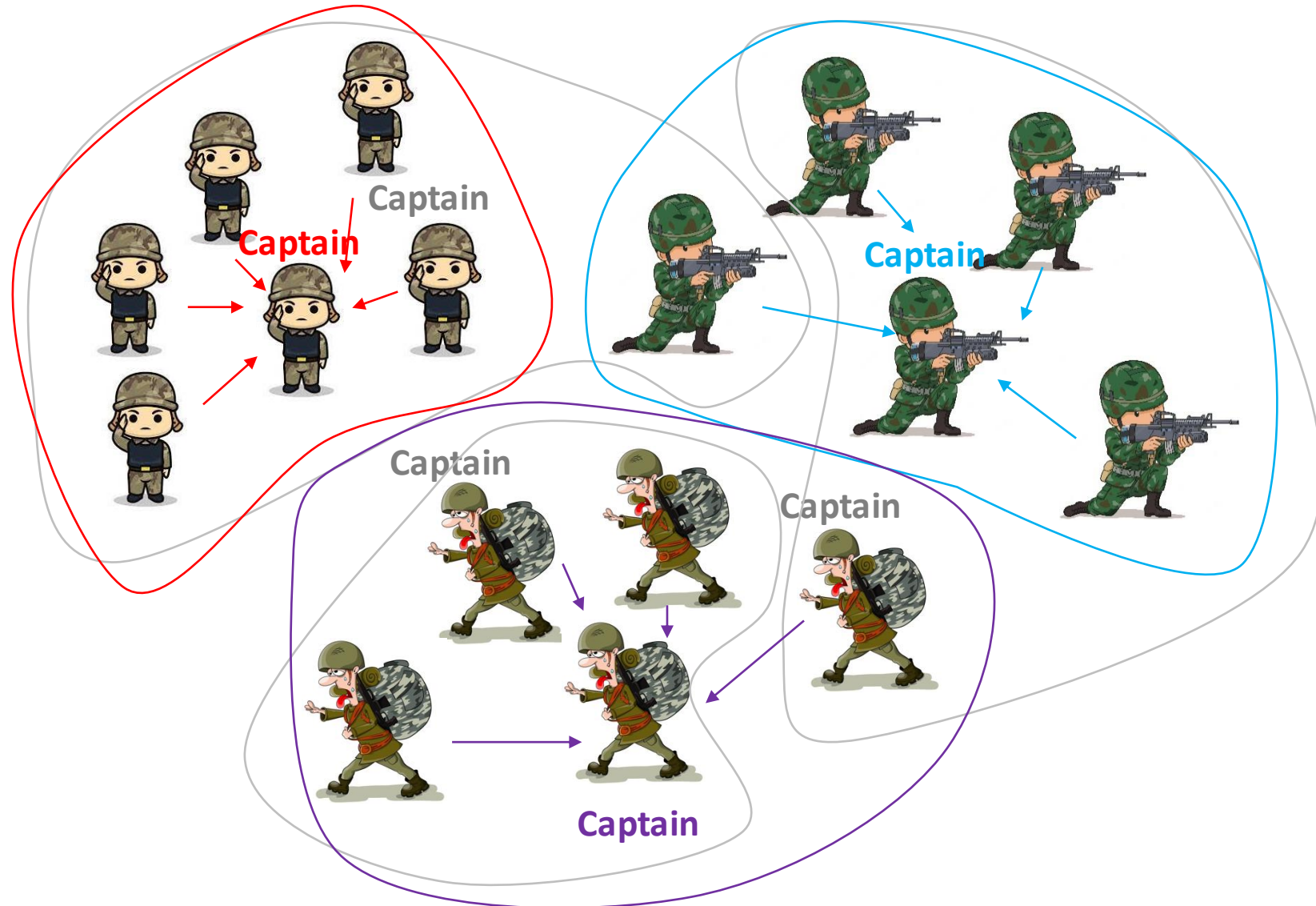


3. Re-election of captains





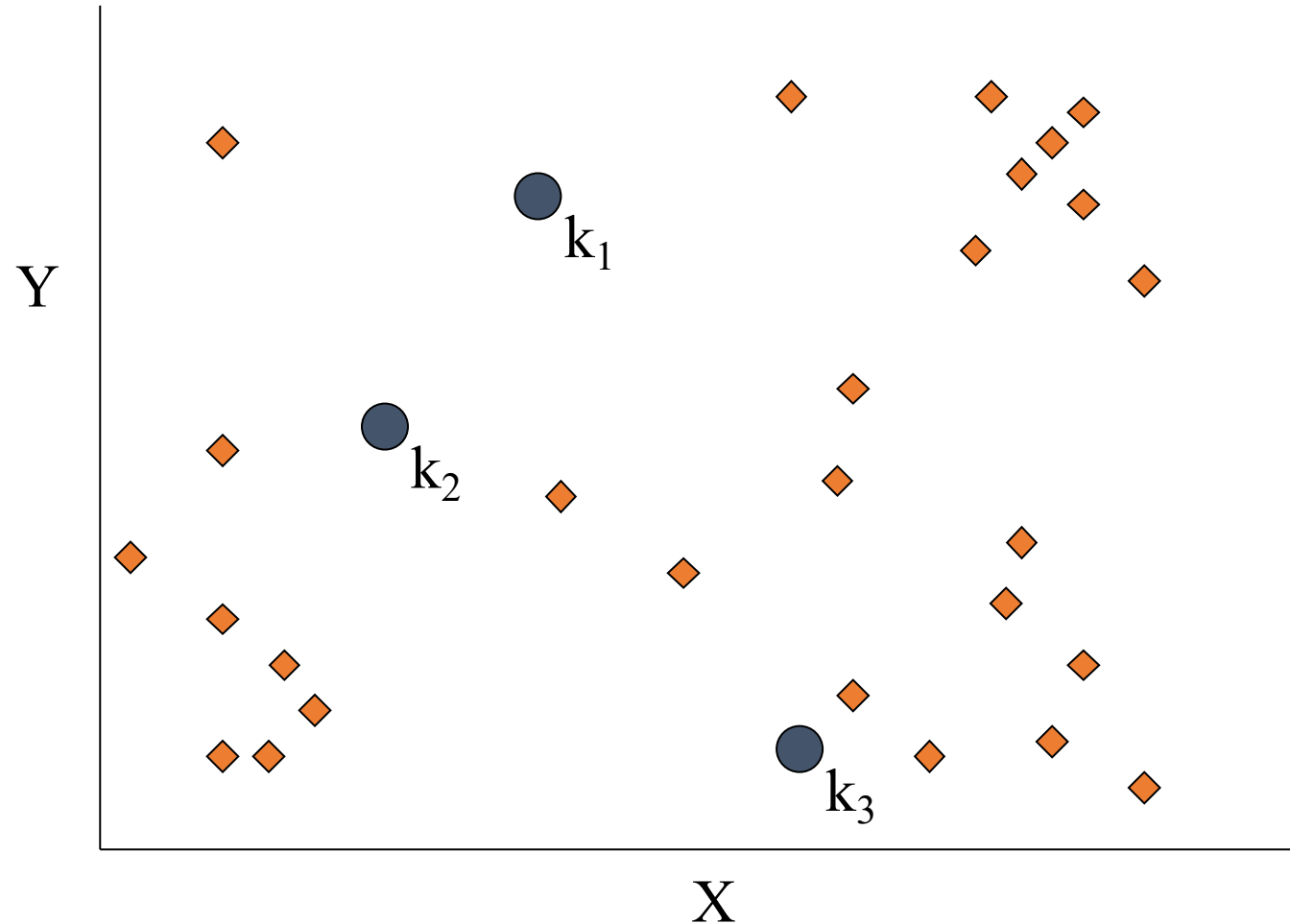
4. Report again





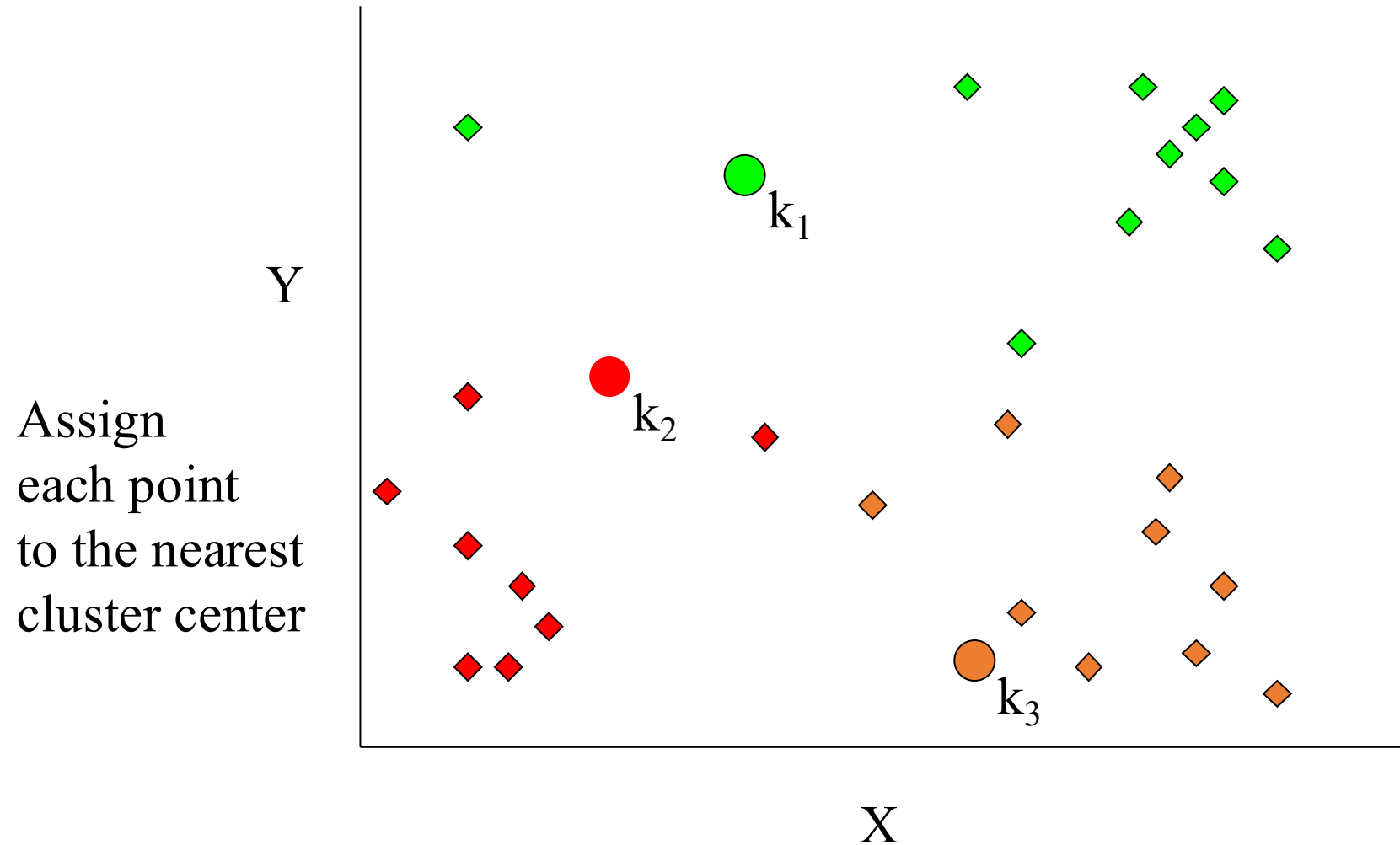
K-Means Step 1

Pick 3
initial
cluster
centers
(randomly)





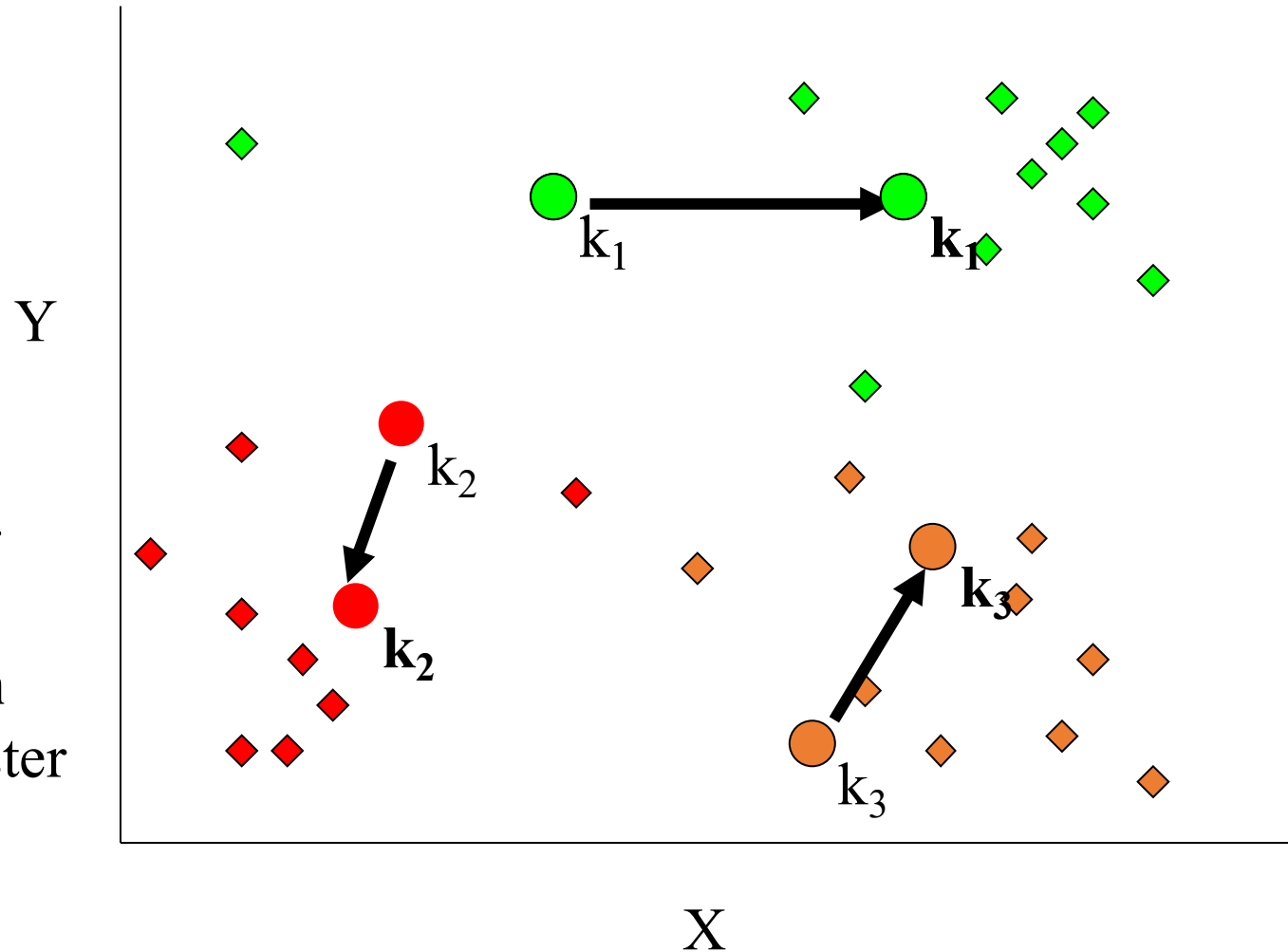
K-Means Step 2





K-Means Step 3

Move
each cluster
center
to the **mean**
of each cluster

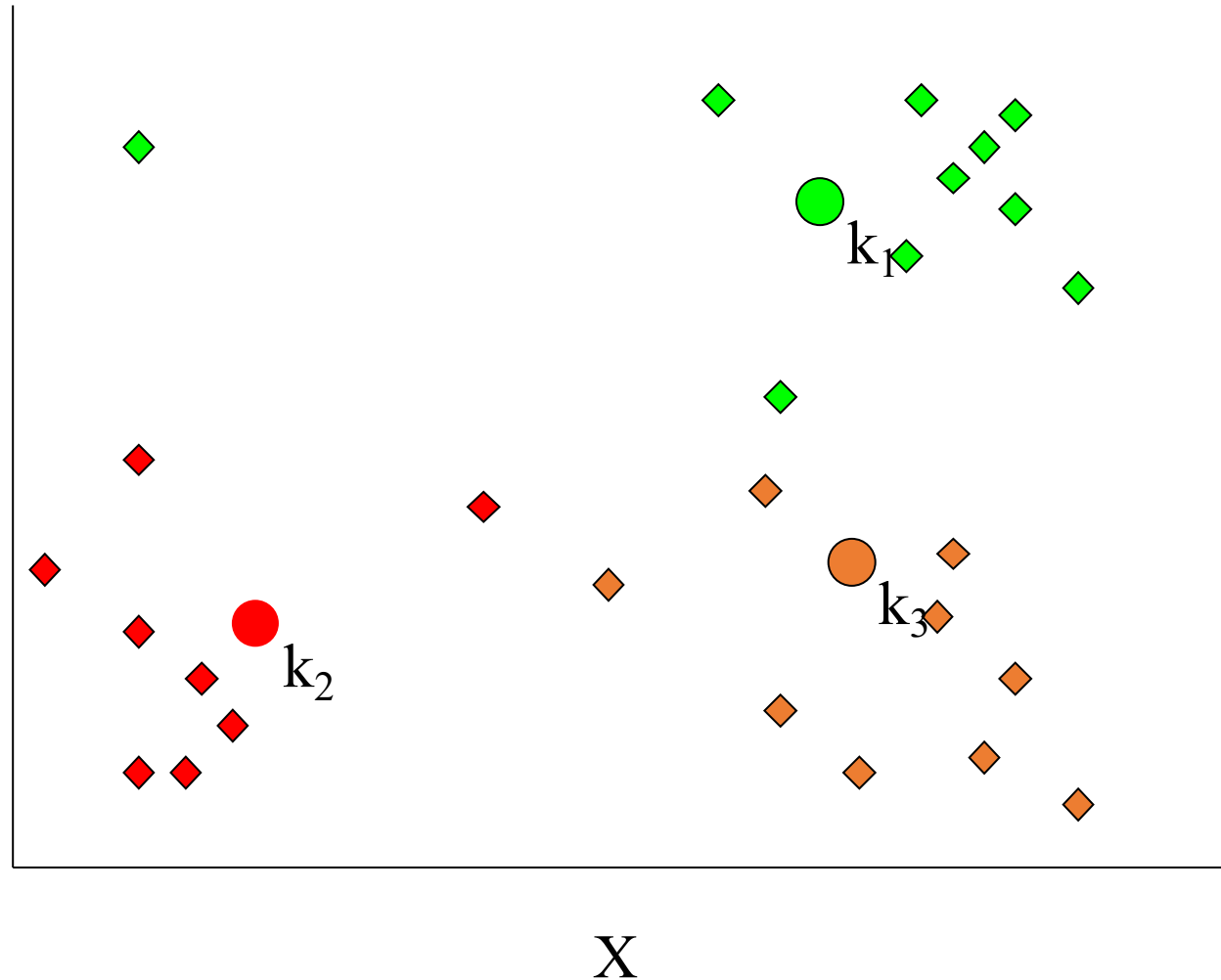




K-Means Step 4

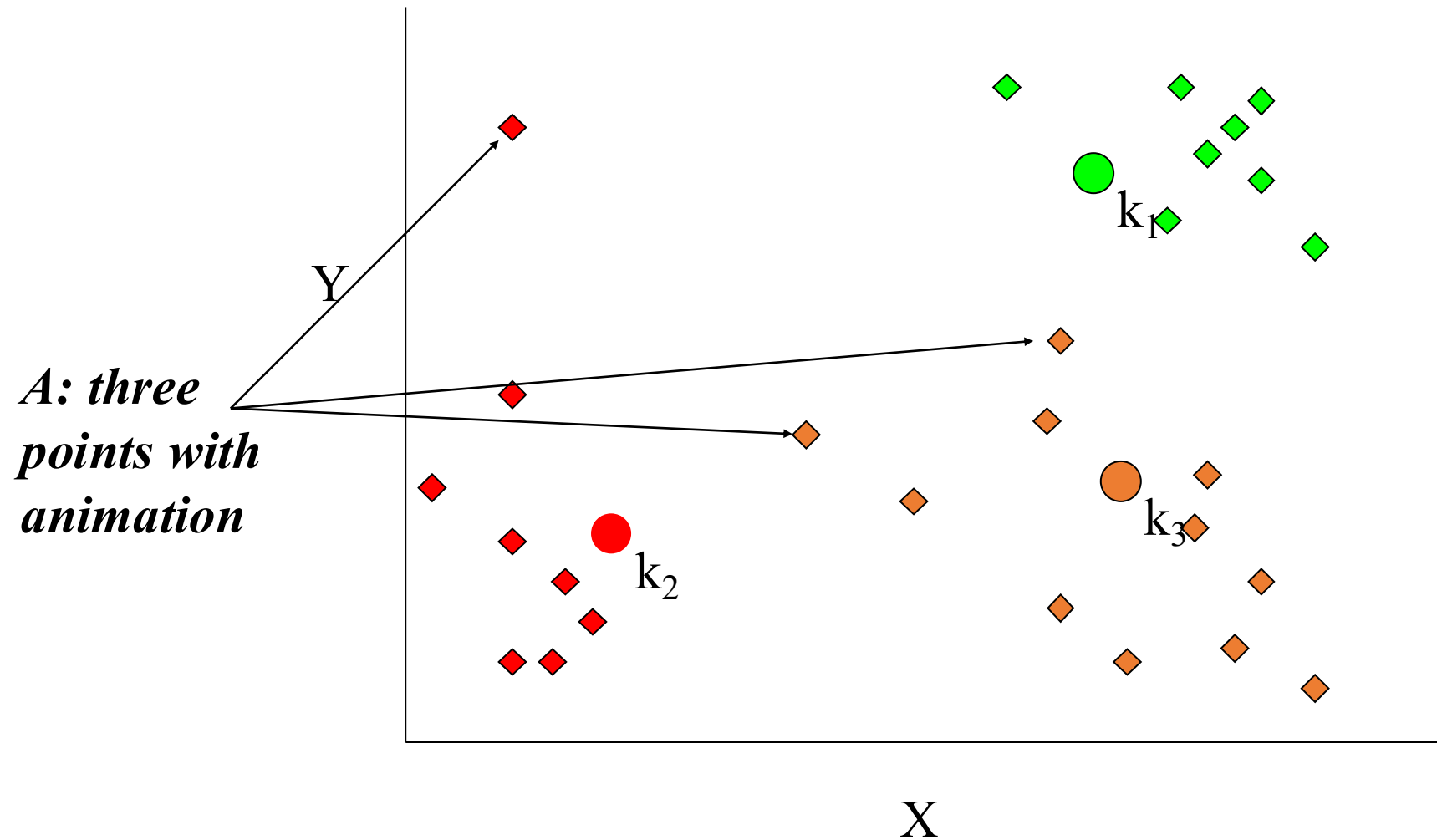
Reassign
points
to the new
(nearest)
cluster center

***Q: Which
points are
reassigned?***



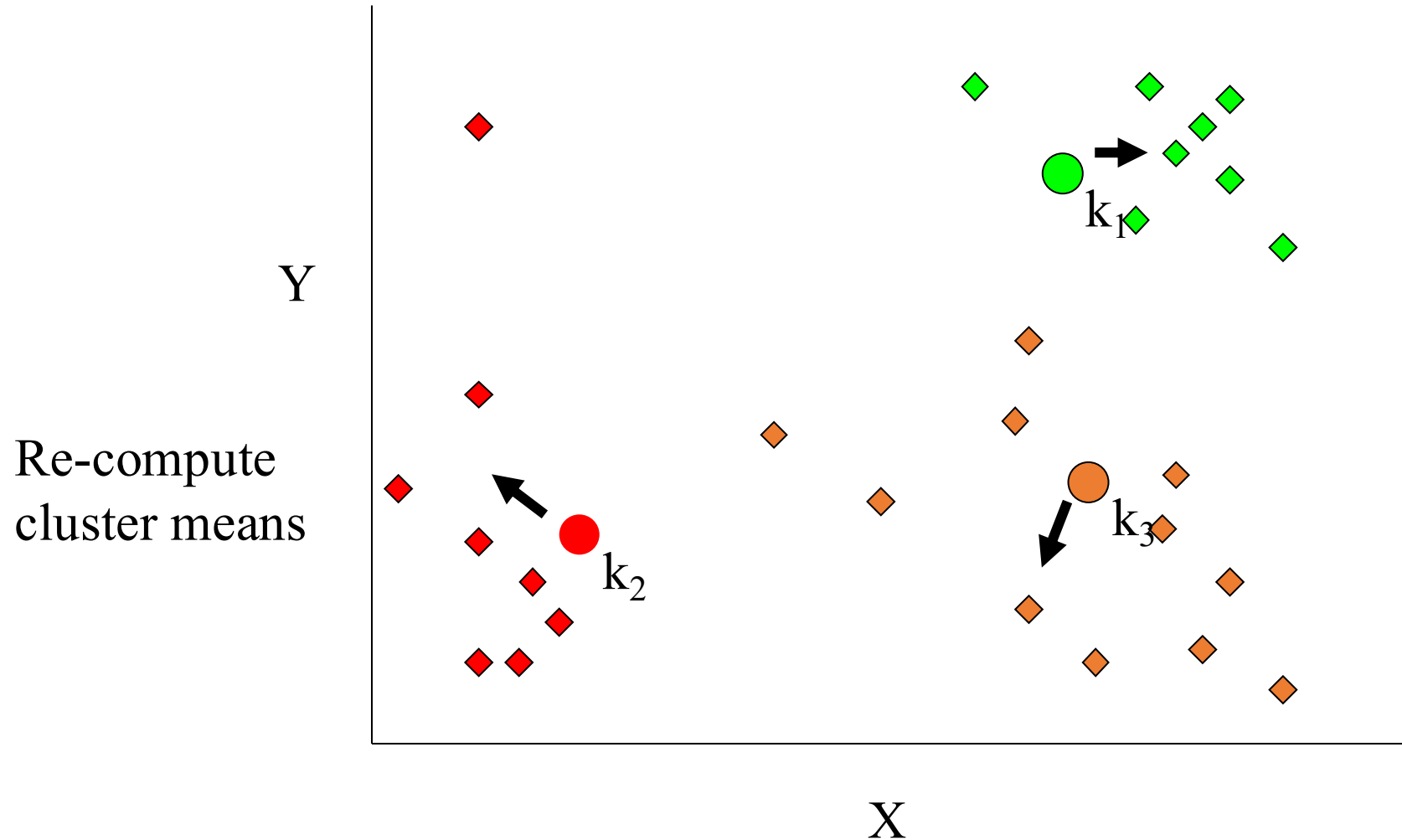


K-Means Step 4 ...



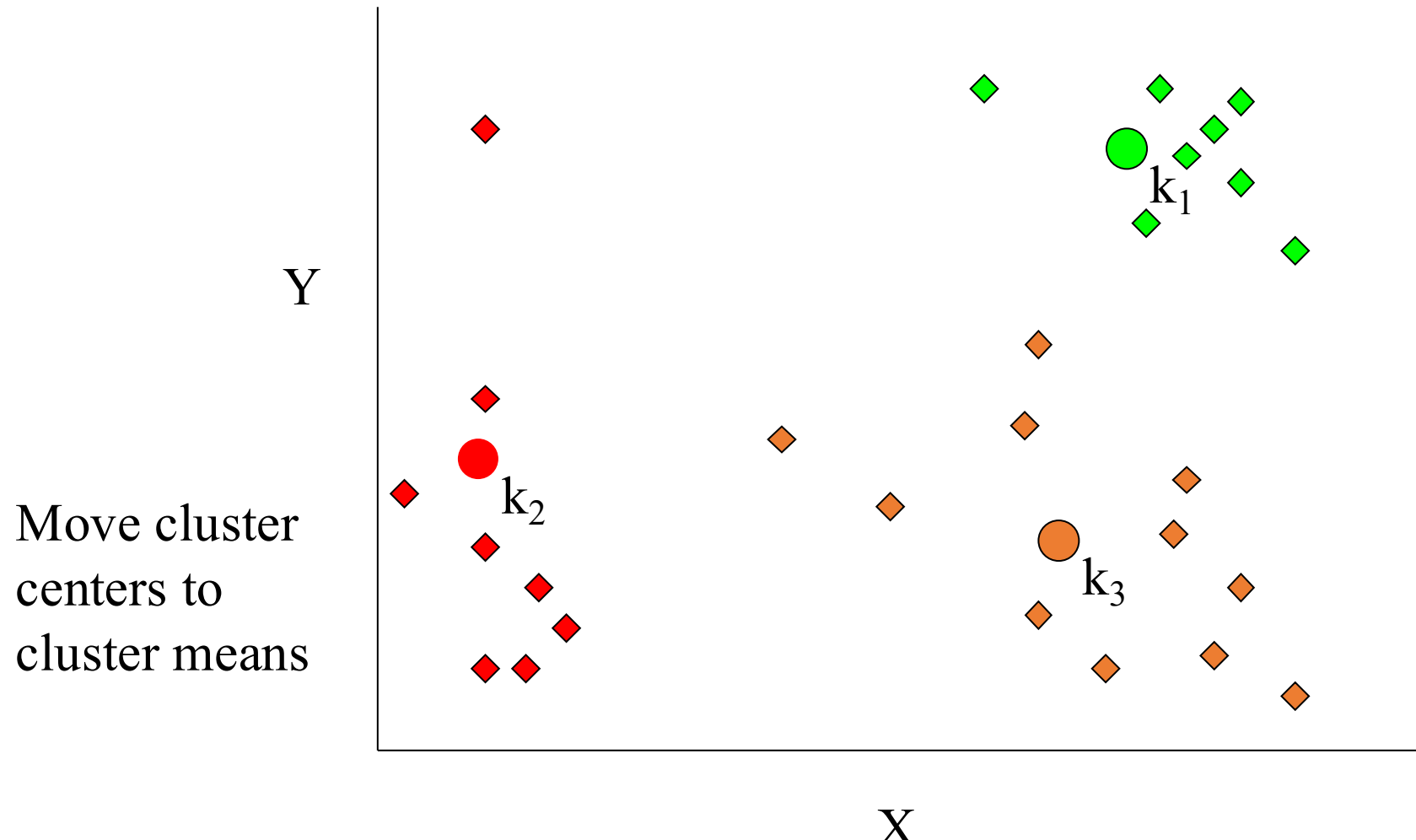


K-Means Step 4 ...



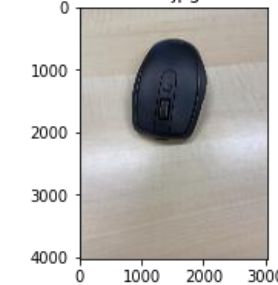
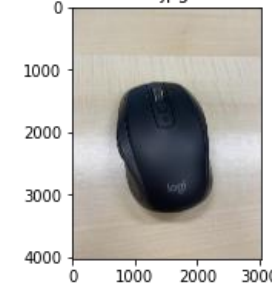
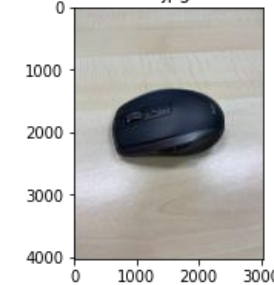
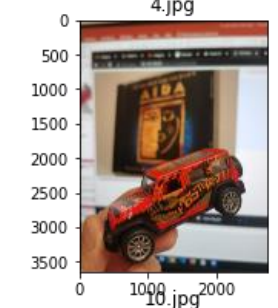
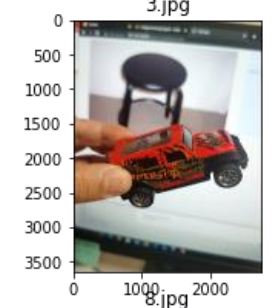
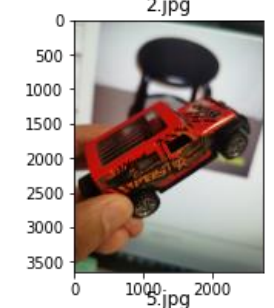
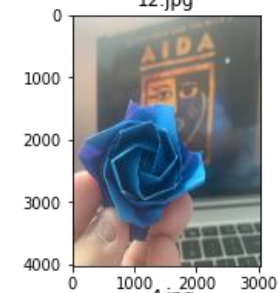
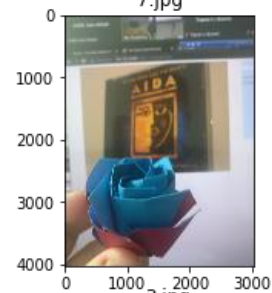
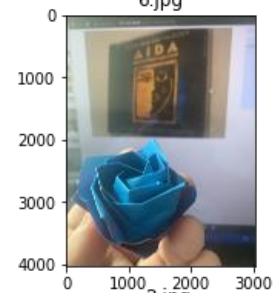
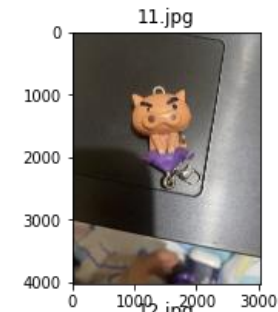
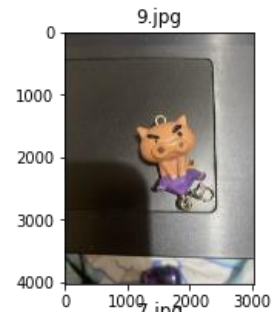
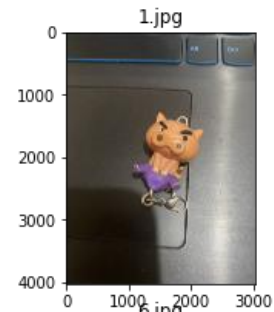


K-Means Step 5



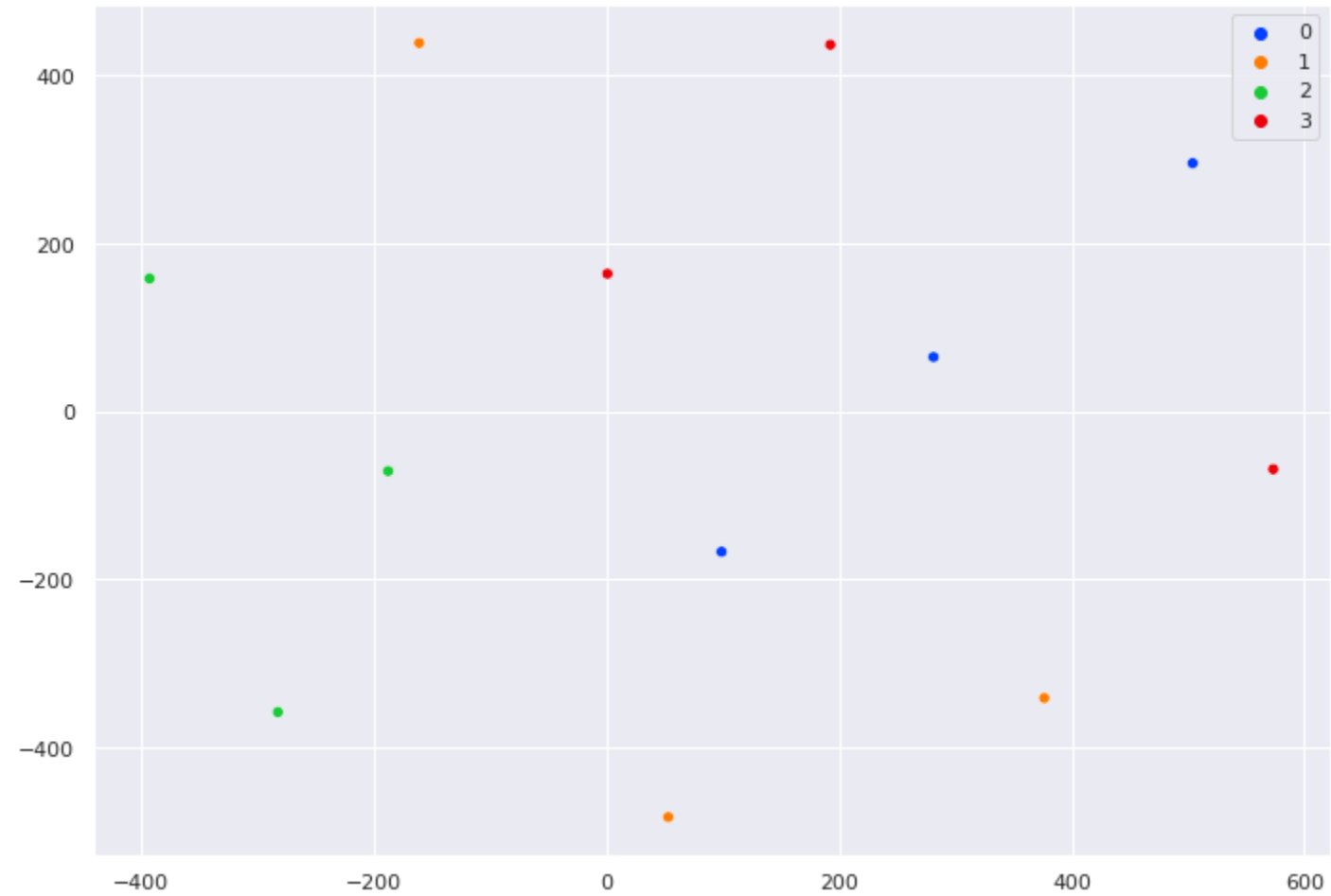


Let's take images from IMHere as examples





Images from IMHere (t-SNE)





Clustering is one of the most representative examples of **Unsupervised Learning**.

We will get back later.

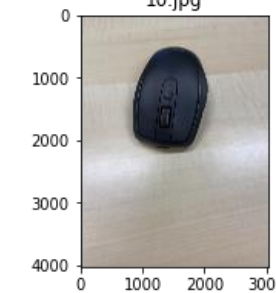
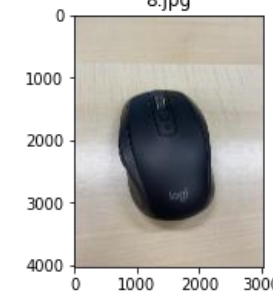
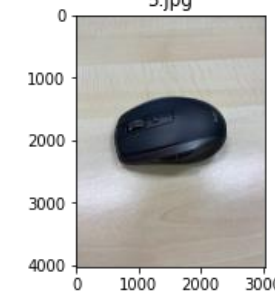
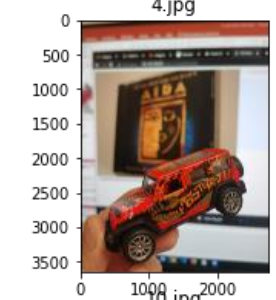
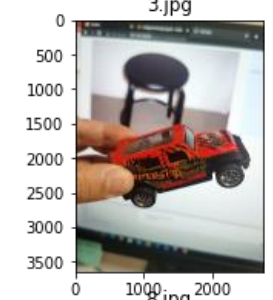
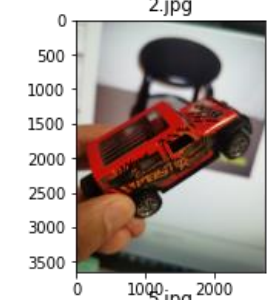
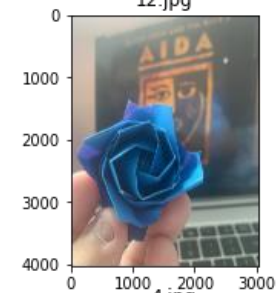
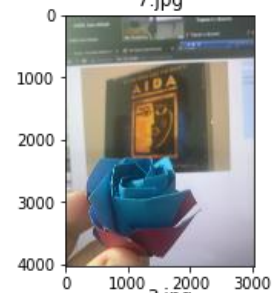
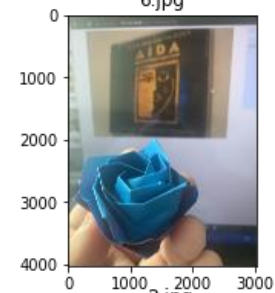
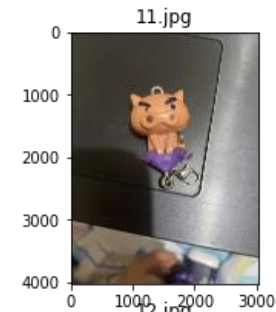
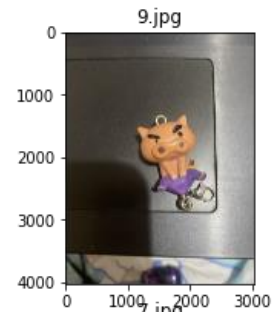
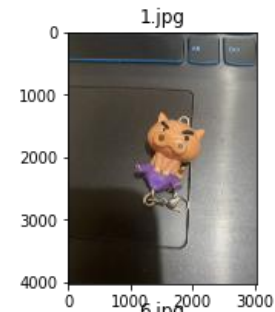


Grouping images is fun.

However, that's not the way we
employed in IMHere.



IMHere – Token-based attendance checking





In IMHere, we are **looking for** images composed of the **similar content** with the one you uploaded. You are checked-in if those are what you used for registrations.



This is called **Content-Based Image Retrieval (CBIR)**.

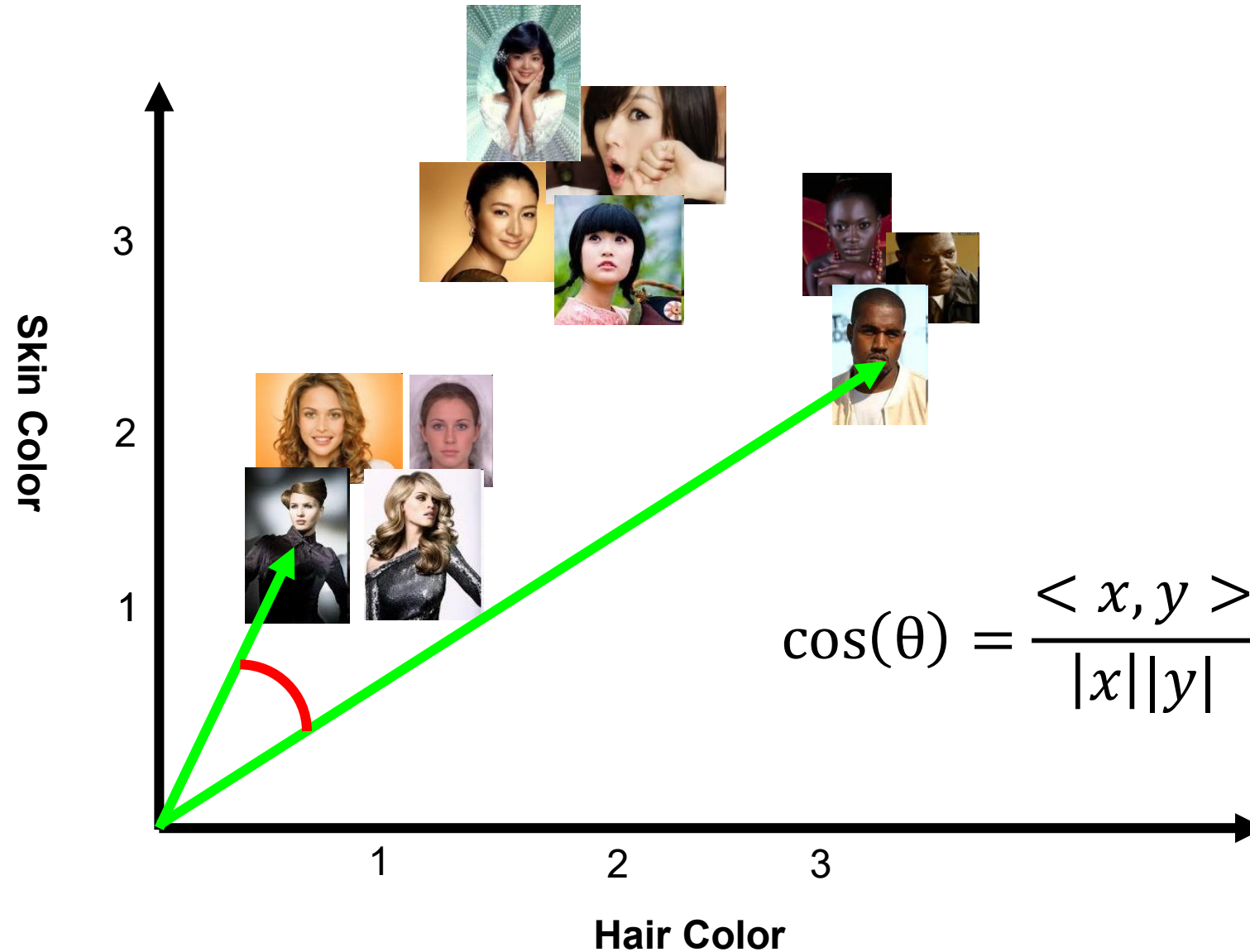


General Steps of CBIR

- > 0. Extract features vectors of all images on file
- > 1. Extract the feature vector for the **query** image
- > 2. Compare it to all the (**target**) images on file by calculating the query-target similarities
- > 3. Sort the similarities in a descending order with which we generate a **ranked list** of the targets
- > 4. Present the ranked list to the searcher

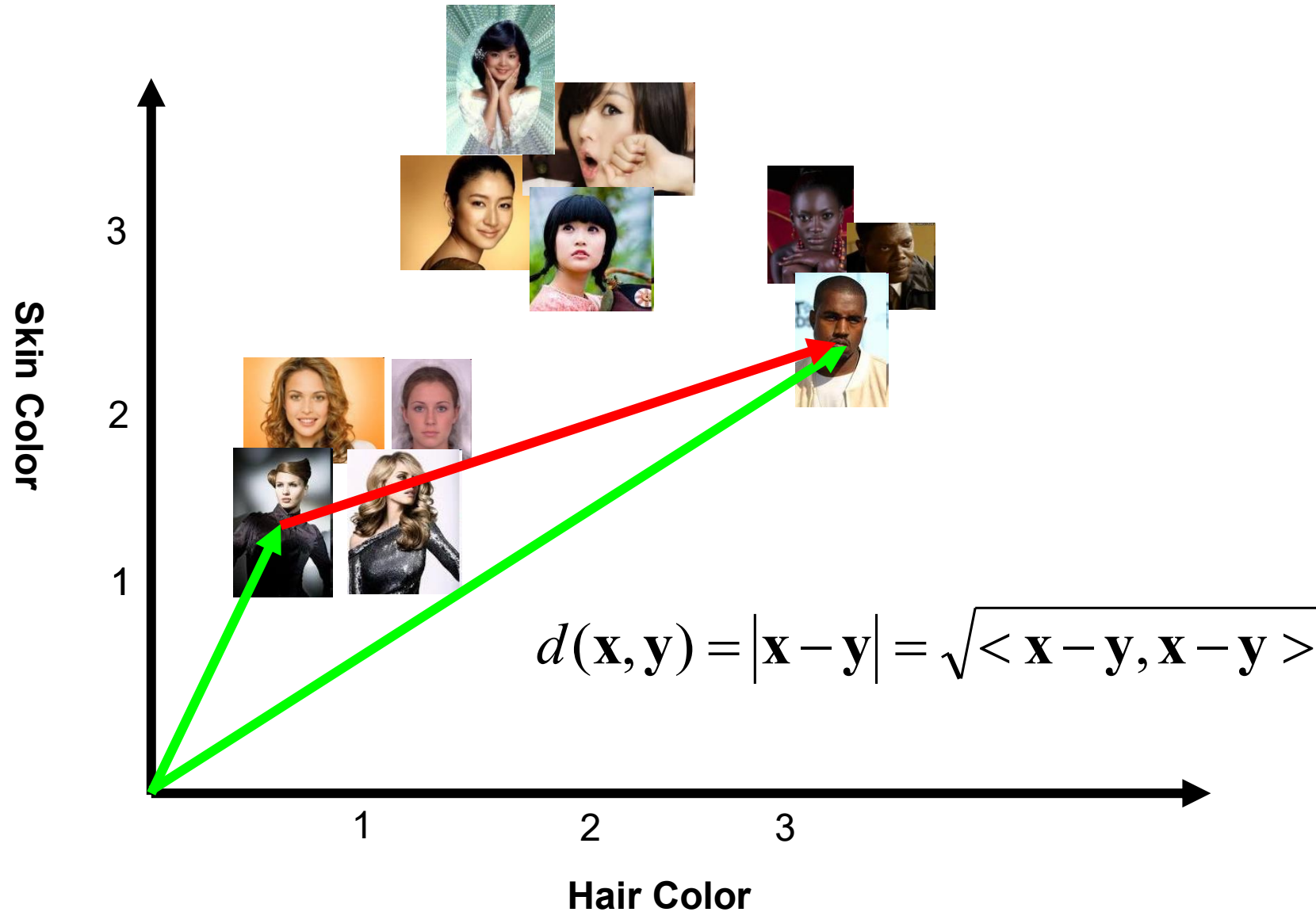


Metrics – Cosine Similarity



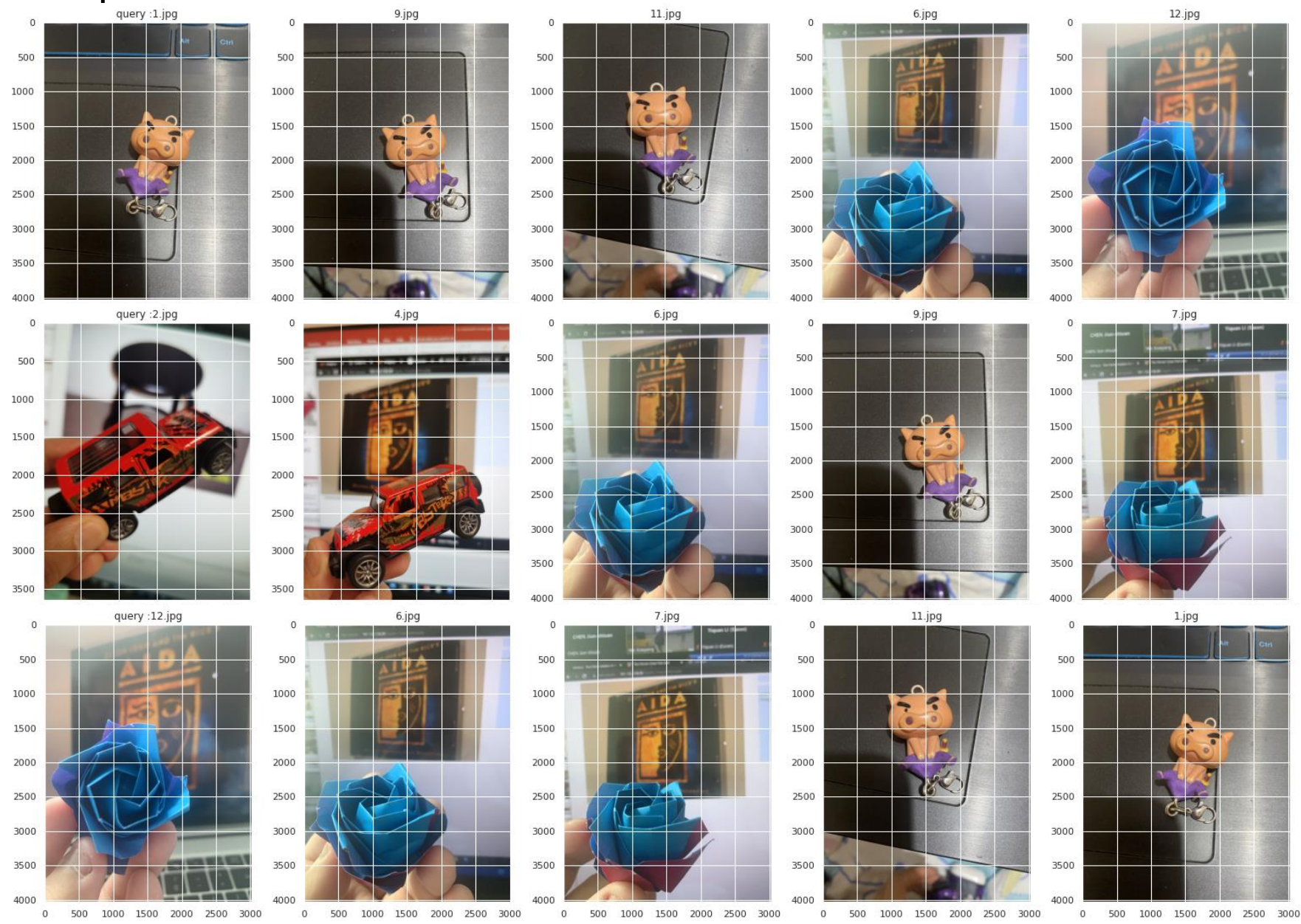


Metrics – Euclidian Distance





Examples from IMHere





It's straightforward to compare the query to the targets in an sequential manner (one by one), but it's not efficient.

Let's assume we have 1 million targets in IMHere. It may take 27 hours per query (100 milliseconds per target).



Is there a better way?



Clustering comes to play!



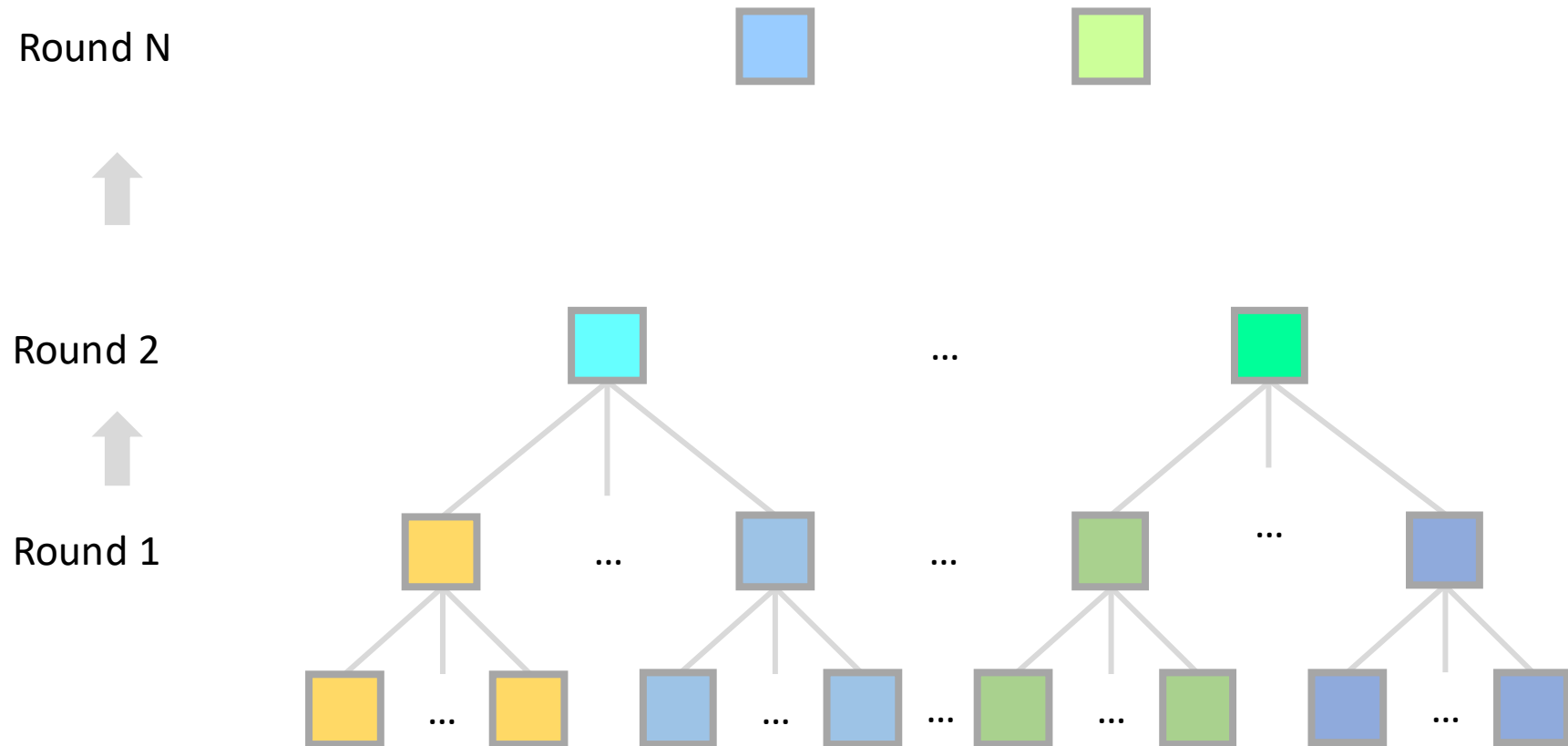
Indexing: group the images as clusters and pick one from each cluster as its representative

Coarse: compare to the representatives only and find top-k

Fine: compare to the member images of the top-k clusters

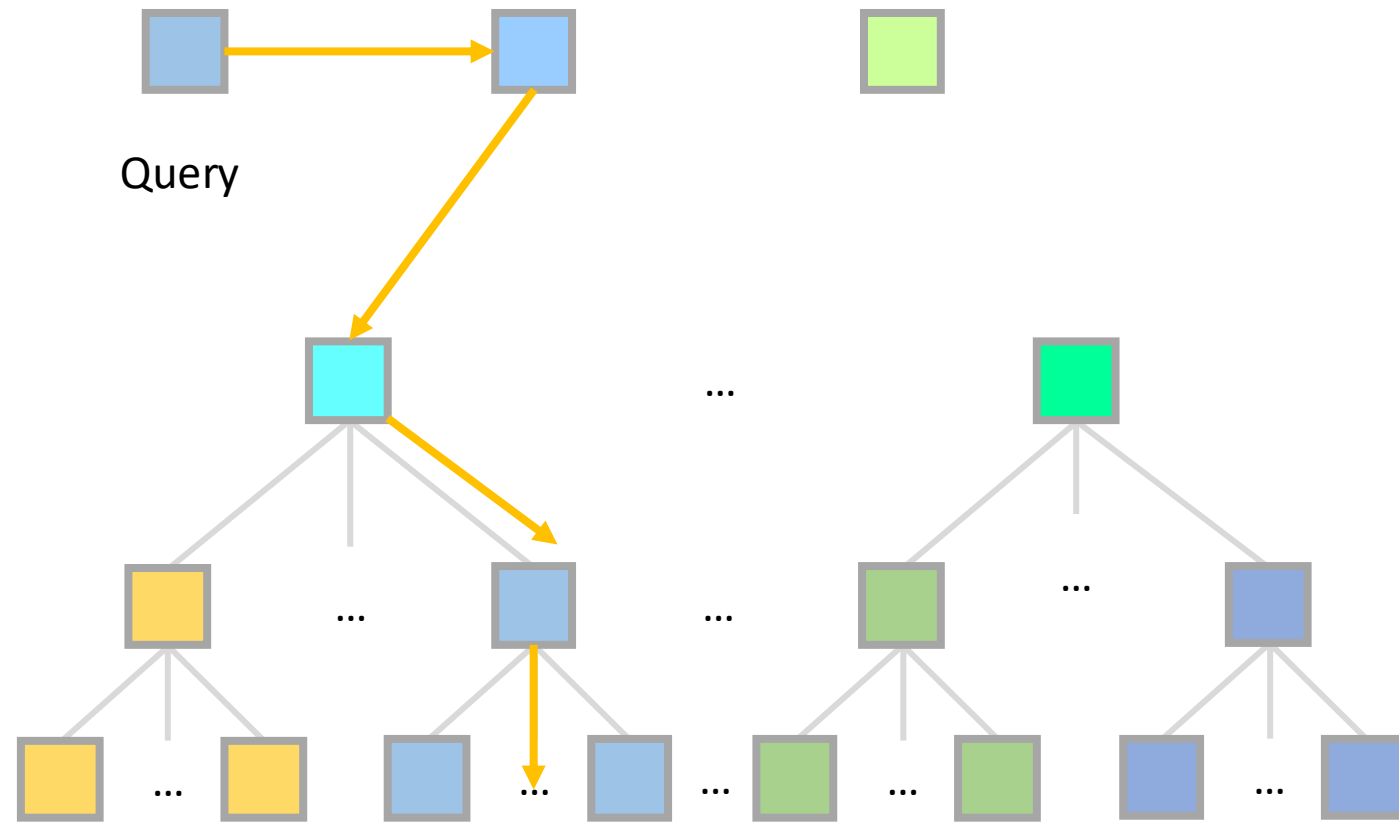


Index the images as a tree



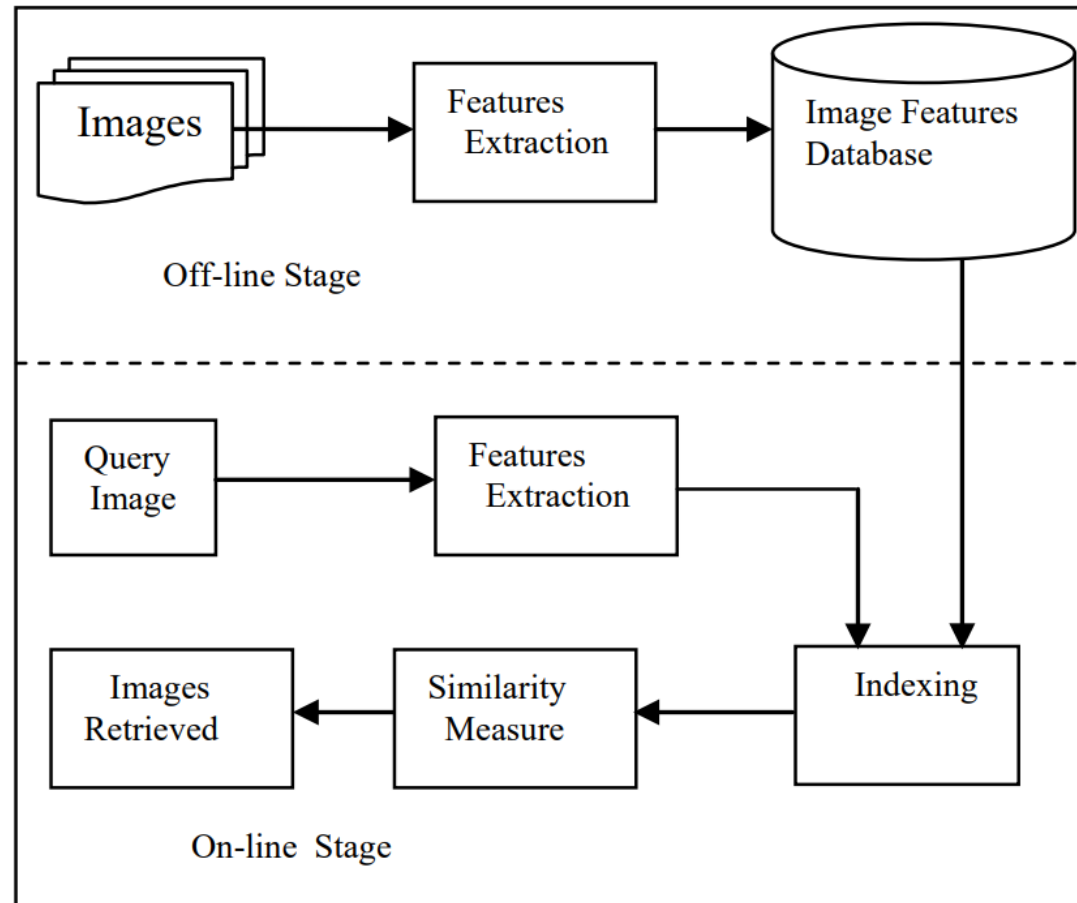


Search from Coarse to Fine





A typical CBIR system



Alkhawlani M, Elmogy M, El Bakry H. Text-based, content-based, and semantic-based image retrievals: A survey[J]. Int. J. Comput. Inf. Technol, 2015, 4(01): 58-66.



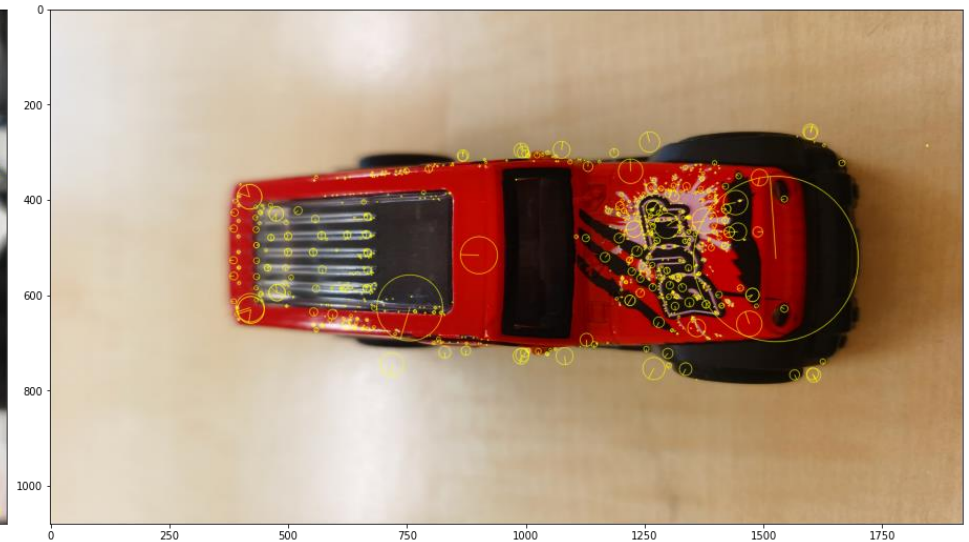
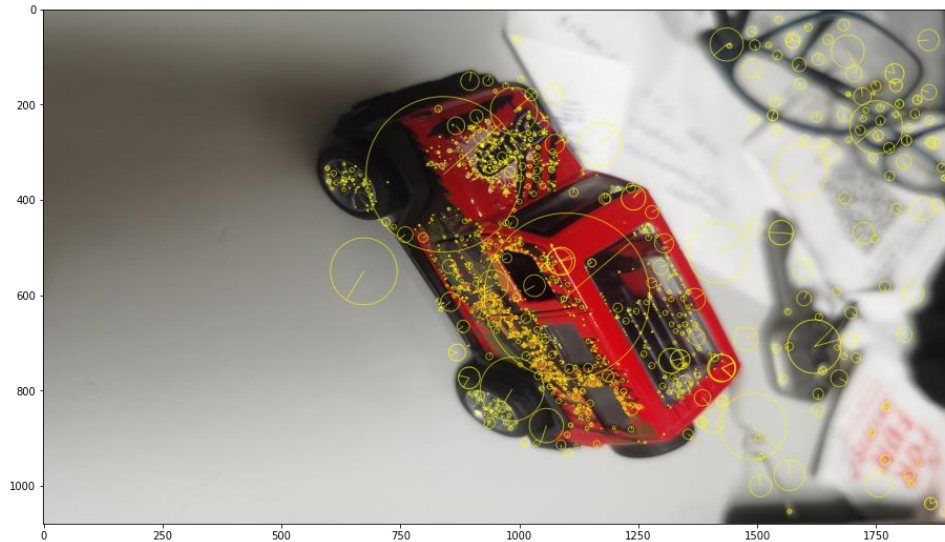
Do you notice?

What we introduced are based on an
assumption of
one-feature-per-image.

How can we deal with images with
multiple feature vectors?



Multiple (Local) Feature Vectors

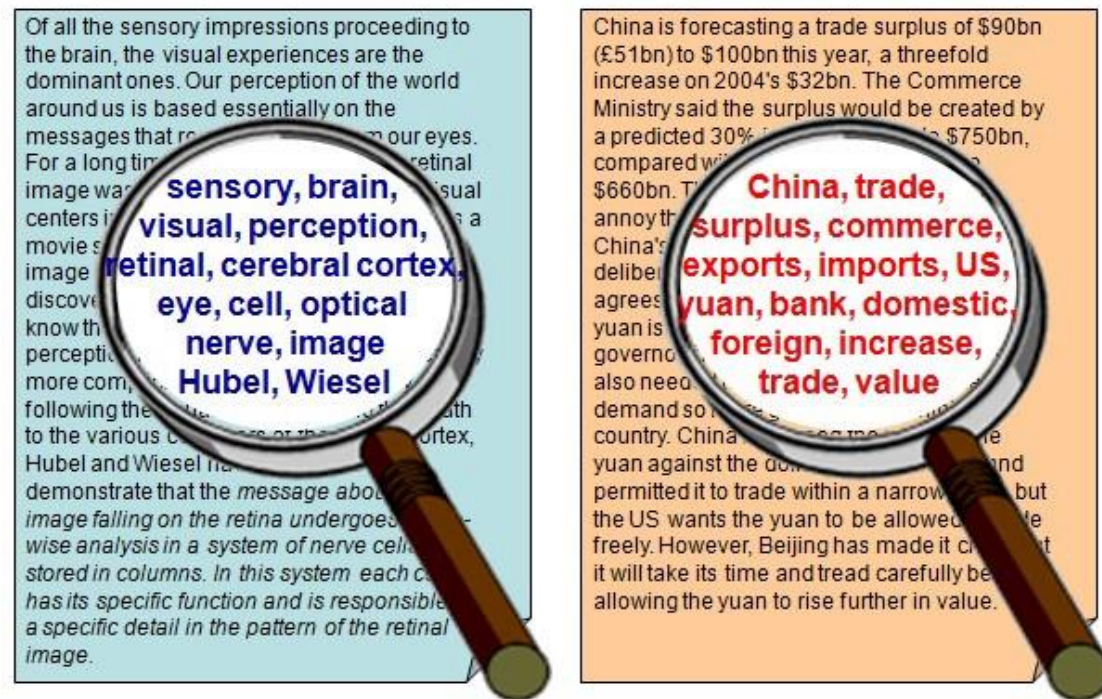




Bag of Visual Words (BoVW)



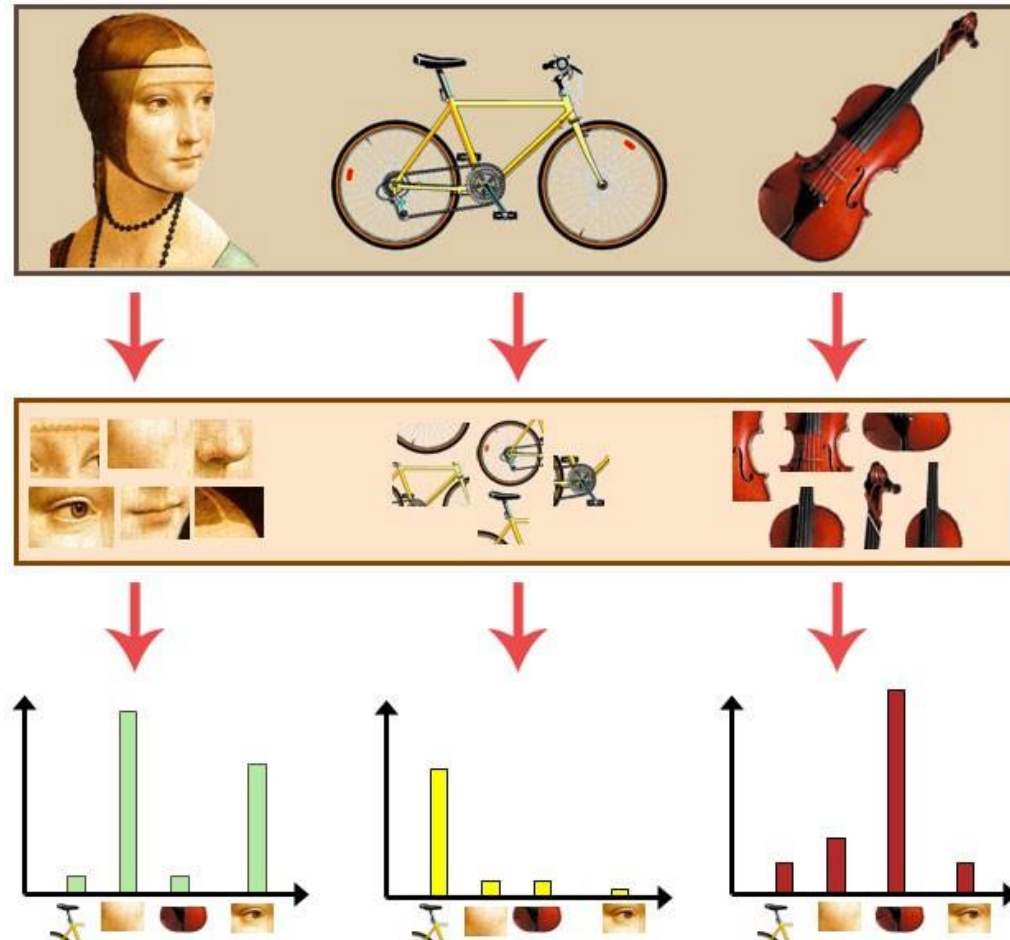
BoW vs. BoVW



Bag of Visual Words in a Nutshell - The art of choosing important features - Bethea Davida

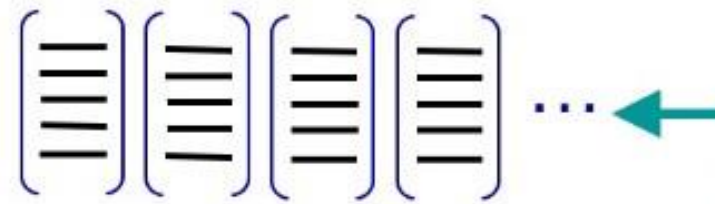


Visual words and bags

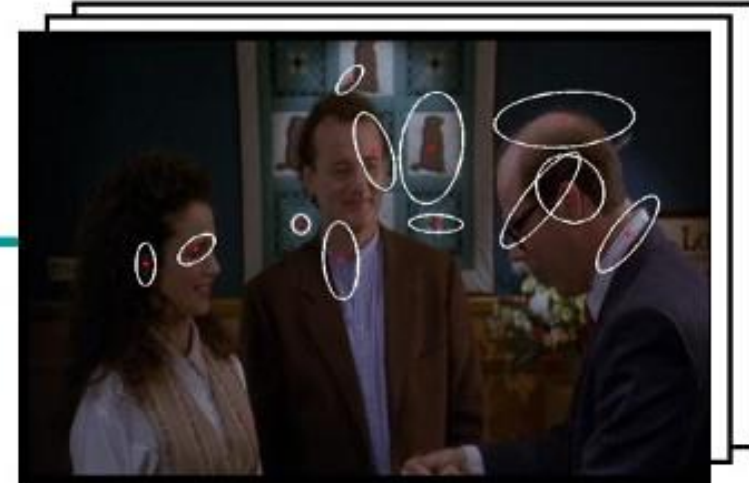




Step 1: Visual Descriptors Extraction



Descriptors



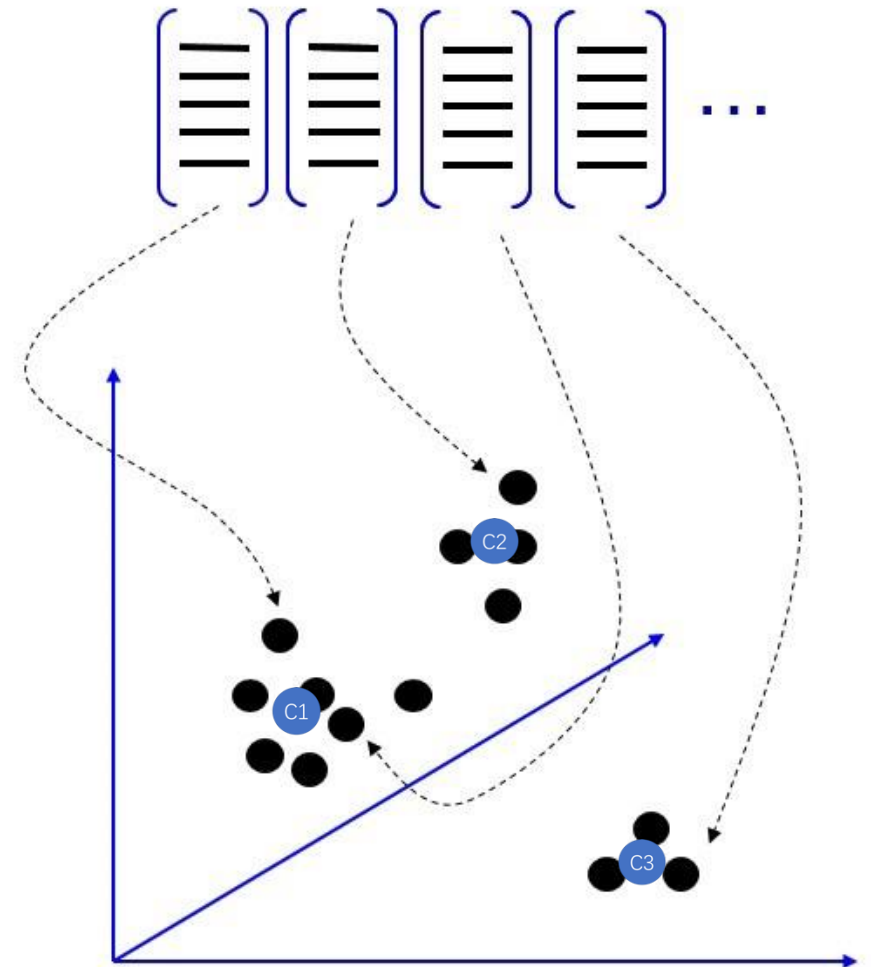


Step 2: Dictionary (CodeBook)

Group the descriptors (from all images) using clustering

Pick one from each cluster as the representative and put them together to construct a dictionary (codebook)

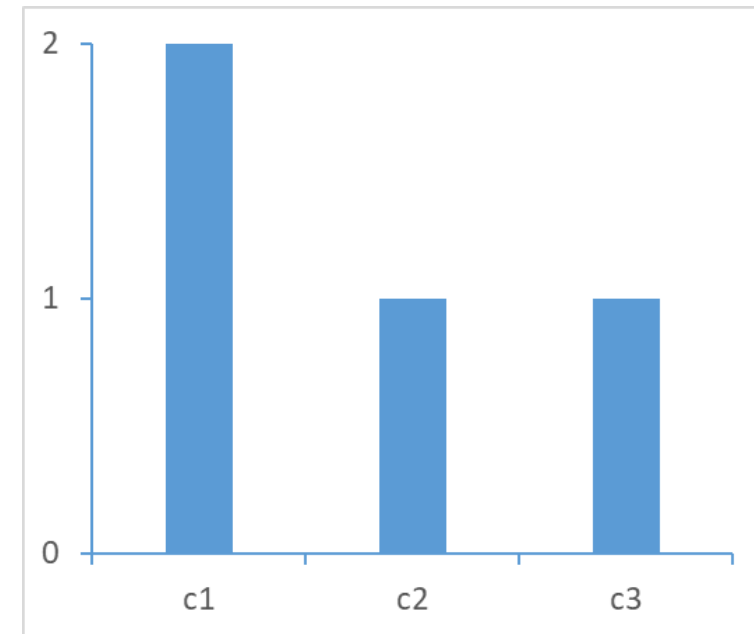
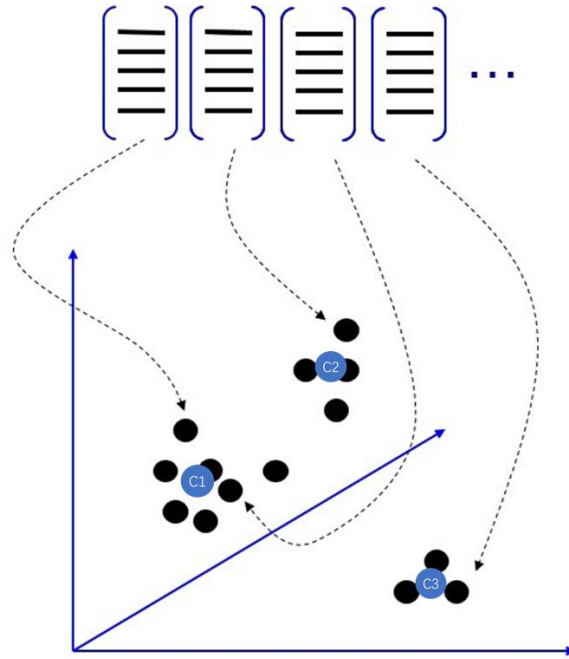
Descriptors in a dictionary are then used as visual words





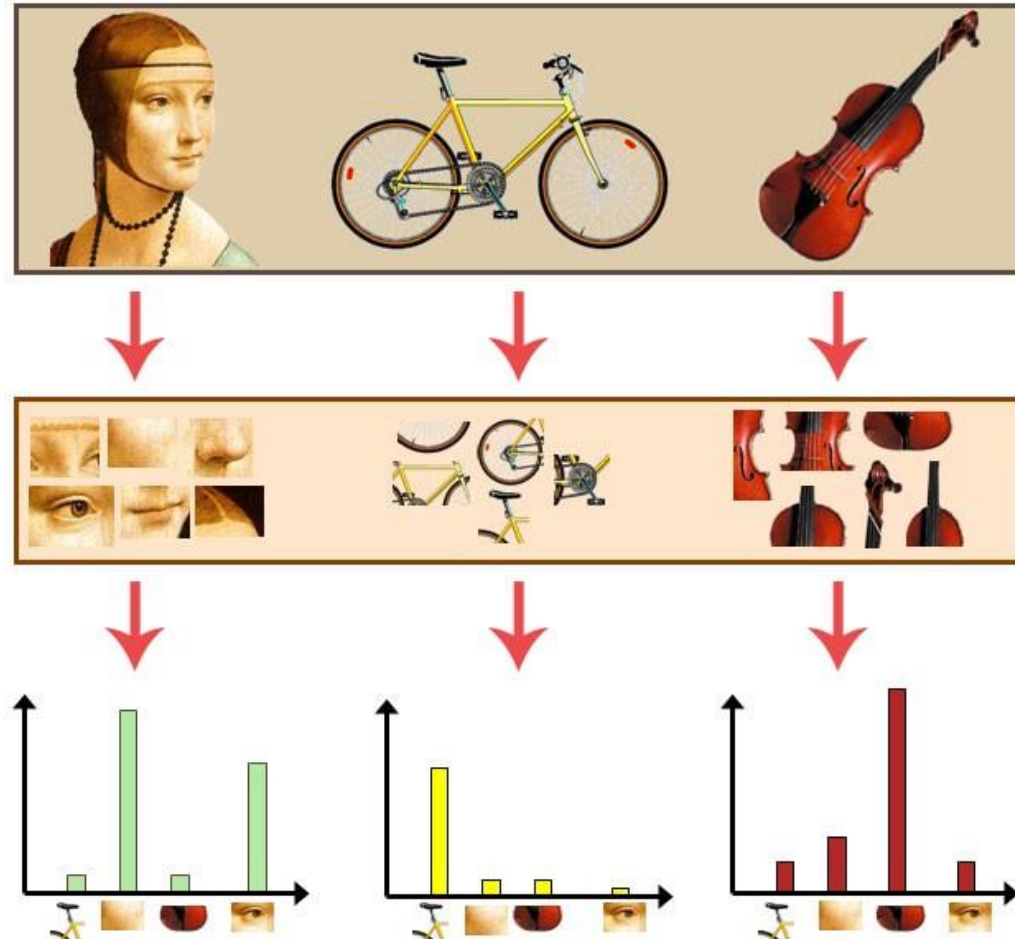
Step 3: Count words in a bag

For each visual word in the dictionary, calculate its frequency in the bag and construct a histogram as the feature vector



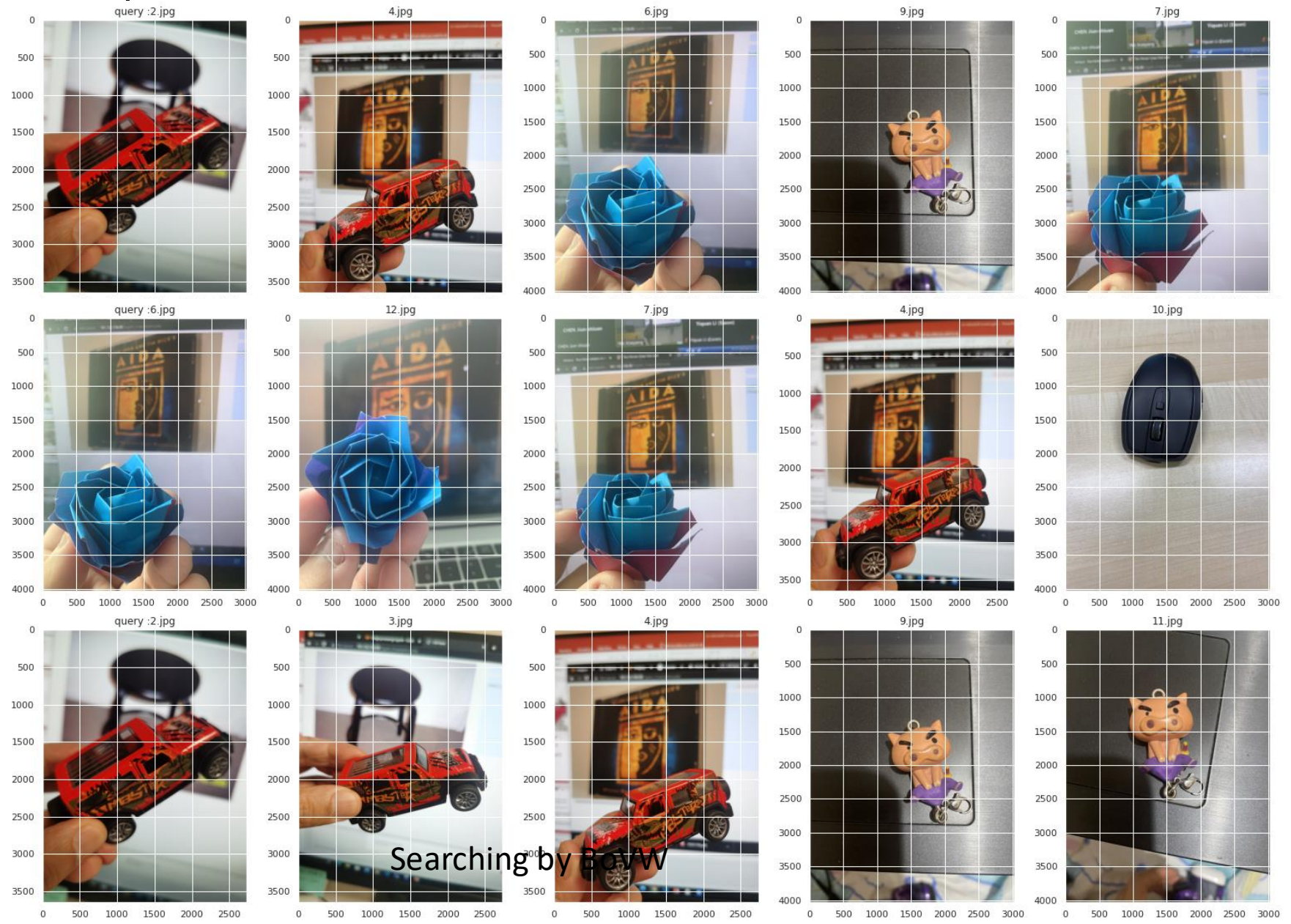


Visual words and bags



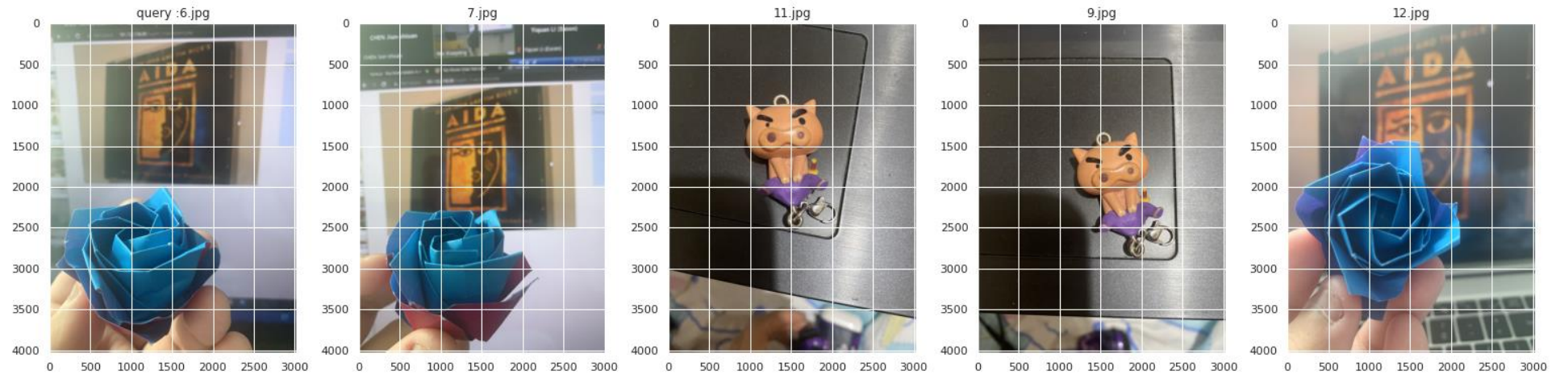


Examples from IMHere

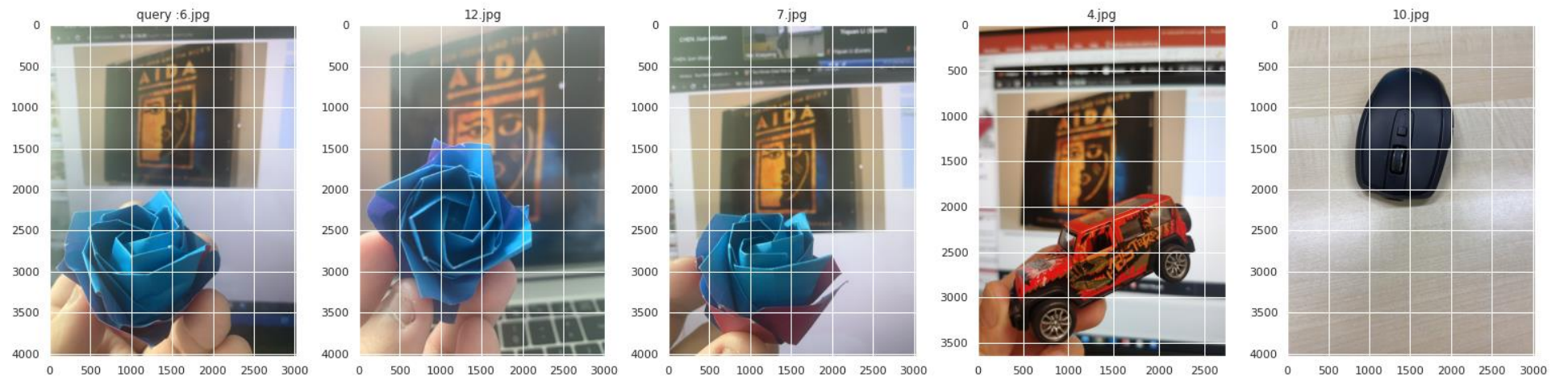




Examples from IMHere



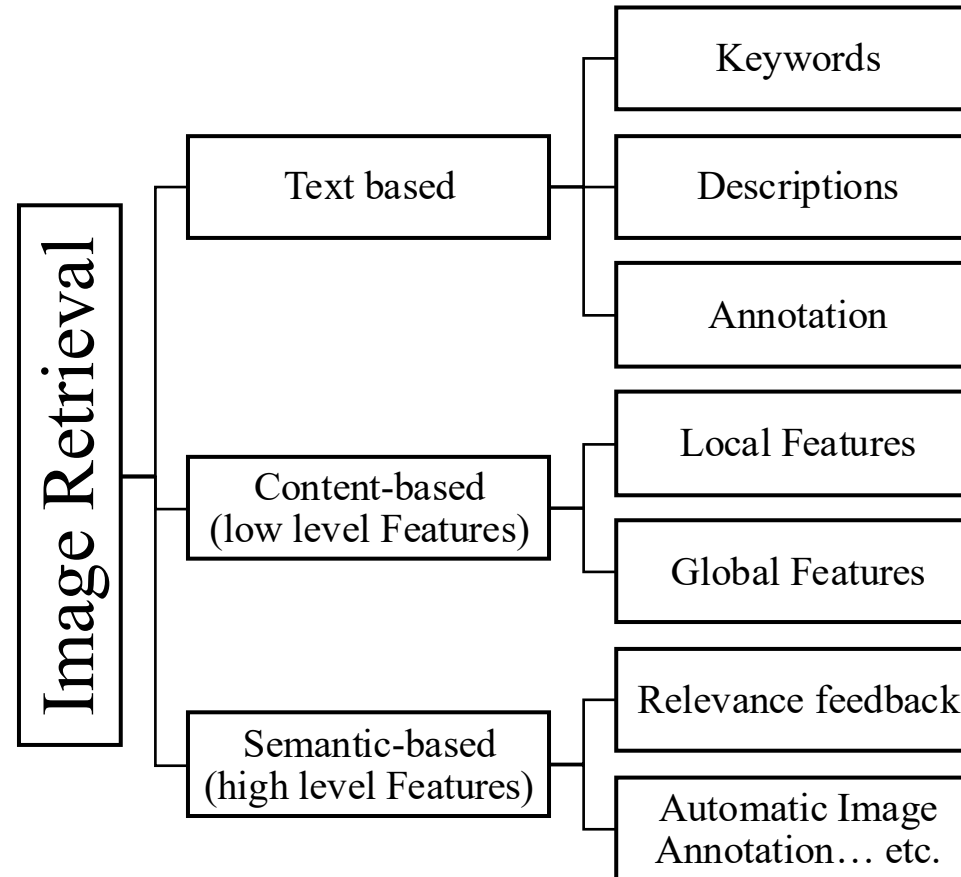
Searching by Histograms



Searching by BoVW



CBIR is not the only solution

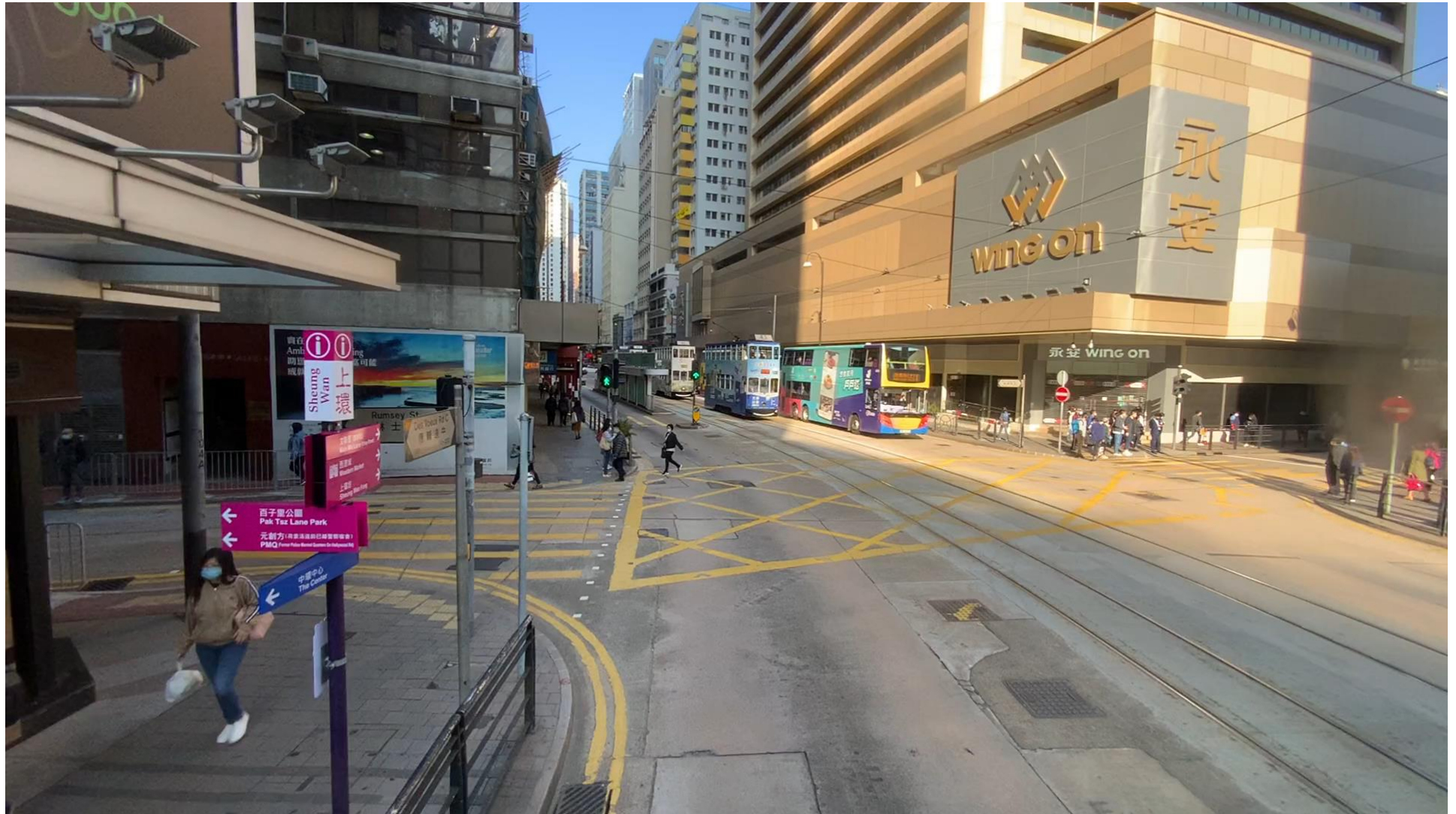




The New Toy



Original Video





The New Toy

- The question to answer: is BoVW able to find the nearest keyframes?

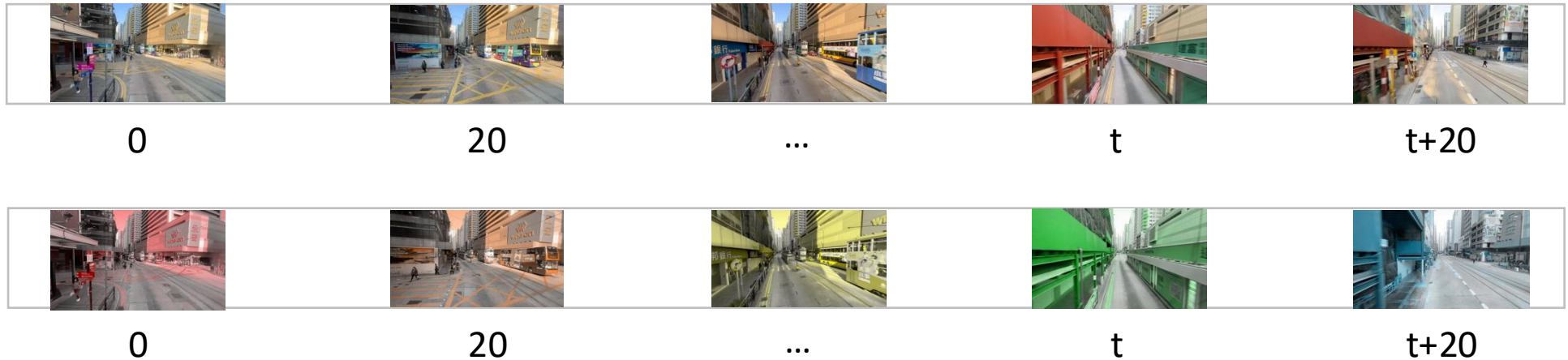


Keyframes selected using a fixed interval of 20 frames



The New Toy

- The question to answer: is BoVW able to find the nearest keyframes?



Keyframe tones modified



The New Toy

- The question to answer: is BoVW able to find the nearest keyframes?



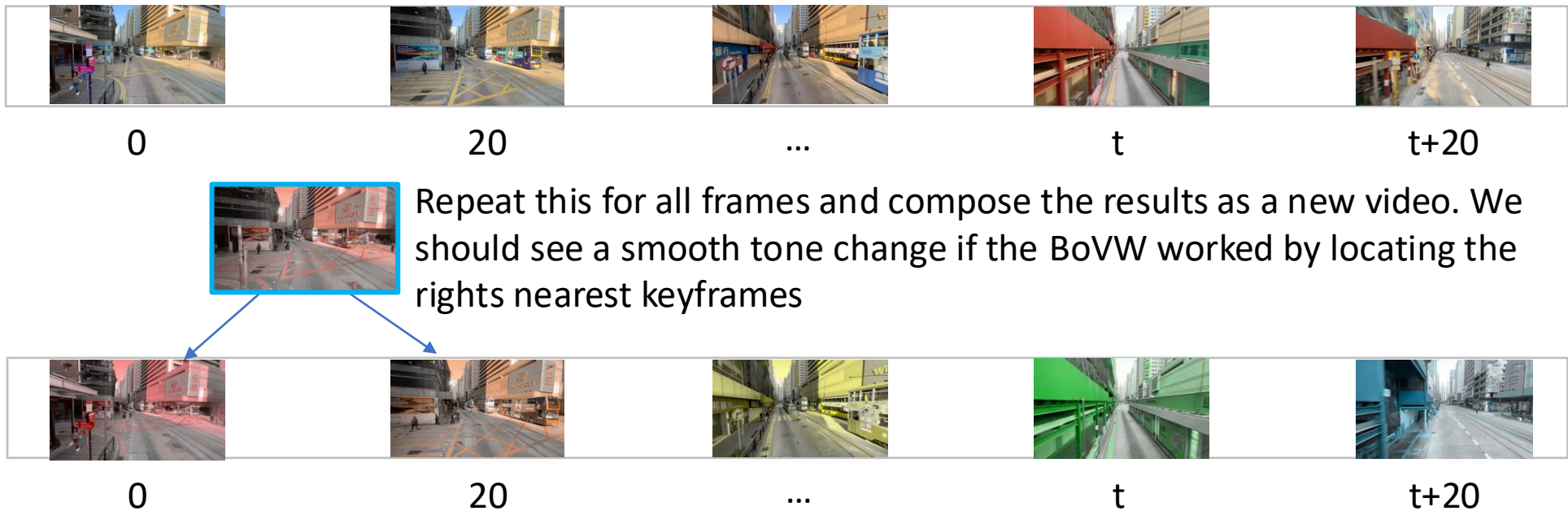
Will the **nearest keyframes** of a query frame in the feature space be the same as its **nearest keyframes in the video**?





The New Toy

- The question to answer: is BoVW able to find the nearest keyframes?





Modified Video





Code to modify the tone of an image (nparray in BRG)

```
def change_tone(frame, shift):  
    # convert the image data to HSV space  
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)  
    h, s, v = cv2.split(hsv)  
    # modify hue channel by adding shift and modulo 180  
    h2 = np.mod(h*0.0 + shift, 180).astype(np.uint8)  
    # convert back to RGB space  
    frame_new = cv2.cvtColor(cv2.merge([h2, s, v]), cv2.COLOR_HSV2BGR)  
    return frame_new
```



Please take it as a challenge and we will release the sample code later. In fact, you will have all the necessary code by completing the tasks of our next tutorial.



TODO: Study following concepts by yourself

1. Dimensionality Reduction (e.g., PCA, MDS, t-SNE)
2. Unsupervised vs. Supervised Learning
3. Density-based Clustering (e.g., DBScan, Mean-Shift)
4. Modern Indexing Structures
5. Data Separability



After-class Reflection and Beyond

1. What is clustering in the context of computer vision?

Describe what it means to group images automatically based on similarity.

2. What do “inter-cluster distance” and “intra-cluster distance” mean?

Why does clustering try to maximize inter-cluster distance and minimize intra-cluster distance?

3. What are the possible weaknesses of K-Means clustering?

Consider issues such as random initialization, choosing the value of K, non-spherical clusters, and sensitivity to outliers.

4. Why is choosing the value of K important in K-Means?

What might happen if K is too small or too large?

5. Why is clustering not always the best solution for the IMHere attendance-checking problem?

Discuss why matching a query token image to registered token images is different from simply grouping all images into clusters.

6. Why can sequential search become inefficient in large-scale image retrieval?

If a database contains millions of images, why is comparing the query with every image one by one impractical?

7. How can clustering help speed up image retrieval?

Explain the idea of using clusters or representatives to avoid comparing the query with every database image.



After-class Reflection and Beyond

8. Why do we need dimensionality reduction methods such as PCA, MDS, or t-SNE?

Explain why high-dimensional feature vectors are difficult to visualize directly.

9. Why should we be careful when interpreting a 2D t-SNE visualization?

Discuss why overlapping points in a 2D projection may not necessarily mean the original high-dimensional features are inseparable.

10. What problem does Bag of Visual Words solve?

Explain why local features such as SIFT create multiple descriptors per image, and why this makes direct image-to-image comparison difficult.

11. What information is lost when BoVW ignores the spatial arrangement of local features?

For example, if two images contain similar visual words but arranged differently, could BoVW confuse them?

12. Why might BoVW perform better than a global color histogram for token matching in IMHere?

Discuss the importance of local distinctive details.



Thank You!